

Optimizing renewable energy and green technologies in distribution systems through stochastic planning of distributed energy resources

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ARTICLE INFO

Keywords:

Renewable energy
Collaborative energy planning
Distributed generations (DGs)
Plug-in electric vehicles (PEVs)
Stationary and mobile battery energy storage systems (SBESSs/MBESSs)
Power Distribution System Optimization

ABSTRACT

This paper introduces a short-term stochastic Mixed Integer Non-Linear Programming (MINLP) multi-objective optimization model designed for collaborative planning. It integrates renewable-based distributed generations (RDGs) like wind and photovoltaic (PV) sources, plug-in electric vehicle parking lots (PEV-PLs), stationary battery energy storage systems (SBESSs), and mobile battery energy storage systems (MBESSs). The main goal is to enhance the penetration of green energy and Plug-in Electric Vehicles (PEVs) within the distribution system (DS). This objective is pursued through the optimization of various planning and operational decision variables, encompassing the number, location, and size of wind DGs, PV DGs, PEV-PLs, and SBESSs, as well as the number, sizing, and monthly transportation schedule of MBESSs, along with the hourly scheduling of charging and discharging profiles for PEVs, SBESSs, and MBESSs. The decision variables are concurrently minimized across three constrained objective functions: total annual expected investment, maintenance and operational costs, power loss, and voltage fluctuation. To validate the efficacy of the collaborative planning model, a 69-bus DS is selected as the case study, with five different configurations proposed: Base case, wind/PV/PEV-PL, wind/PV/PEV-PL/SBESS, wind/PV/PEV-PL/MBESS, and wind/PV/PEV-PL/SBESS/MBESS. The research findings, summarized in the conclusion sections, highlight the advantages and disadvantages of each proposed configuration regarding economic, environmental, and technical objectives and their impact on green energy and PEV penetration levels.

1. Introduction

1.1. Motivation

A profound global shift in attention towards environmental preservation has been catalysed by the widespread pollution stemming from traditional fuel-powered generators. The foremost objective now is to reduce reliance on fossil fuels and advocate for the widespread implementation of renewable energy sources (RESs) through distributed generations (DGs), with particular emphasis on wind and solar photovoltaic (PV) systems. Nonetheless, a notable challenge persists in the erratic and intermittent characteristics of RESs. This inherent unpredictability injects further uncertainty into the system, rendering precise forecasting challenging and necessitating robust solutions [1–3].

Additionally, the swift proliferation of electric vehicles (EVs) as part of initiatives to mitigate the detrimental impacts of pollution represents a significant advancement in the automotive industry. [4–6]. Projections from the International Energy Agency forecast an astonishing 130 million EVs globally by 2030 [7]. Central to this paradigm shift is the establishment of efficient infrastructure, particularly plug-in electric vehicle parking lots (PEV-PLs). These facilities, outfitted with robust communication links, rapid chargers, controllers, and management systems, are vital hubs facilitating seamless interactions between EV owners and distribution system (DS) operators. They optimize operations both economically and technically. Moreover, incentivizing the adoption of bidirectional EVs with vehicle-to-grid (V2G) capabilities transforms PEVs into supplementary mobile storage assets within the system. By harnessing these capabilities, PEVs can effectively mitigate the variability of RESs, storing surplus energy during periods of

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<https://doi.org/10.1016/j.ecmx.2024.100834>

Received 6 September 2024; Received in revised form 4 December 2024; Accepted 10 December 2024

Available online 17 December 2024

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Nomenclature

Sets

- $\Phi_M, \Phi_D, \Phi_S, \Phi_T$ The sets encompass all months, days, obtained 20 scenarios, and hours
- Φ_{bus} The set includes all buses
- Φ_{brch} The set includes all branches
- Φ_{CL}, Φ_{RL} The set includes all commercial and residential loads
- $\Phi_W, \Phi_{PV}, \Phi_{PEV-PL}, \Phi_{SBESS}, \Phi_{MBESS}$ The set includes all deployed wind DGs, PV DGs, PEV-PLs, and MBESSs
- Φ_{PEV} The set includes the entire fleet of PEVs
- N_{bus} The number of buses
- $\epsilon_{s,d,m}$ The joint probabilities were obtained using BRM

Decision and binary variables

- $\delta_{(W/PV)/(PEV-PL)/SBESS/MBESS}^{number}$ The positive integer output decision variable represents the assigned numbers of wind DGs, PV DGs, PEV-PLs, SBESS, and MBESSs
- $N_{(W/PV)/(PEV-PL)/SBESS/MBESS}^{max}$ The maximum allowable numbers of wind DGs, PV DGs, PEV-PLs, SBESS, and MBESSs
- $N_{Wu/PVU/(PEV-PL)/SBattu/MBattu}^{max}$ The maximum allowable number of internal units, including wind turbines, PV units, PEVs, and stationary and mobile battery units
- $\delta_{j,(W/PV)/(PEV-PL)/SBESS}^{Location}$ The positive integer output decision variable represents the assigned locations of wind DGs, PV DGs, SBESSs, and PEV-PLs
- $\delta_{j,m,MBESS}^{Location}$ The positive integer output decision variable denotes the monthly scheduled location of MBESS
- $\delta_{j,(W/PV)/(PEV-PL)/SBESS/MBESS}^{size}$ The positive integer output decision variable represents the internal units for each wind DG, PV DG, PEV-PL, SBESS, and MBESS
- $\delta_{j,t,s,d,m}^{status,(SBESS,MBESS)}$ The bounded integer output decision variable represents each SBESS and MBESS's operating status (charging, discharging, and floating)
- $\delta_{i,t,s,d,m}^{ch,PEV}, \delta_{i,t,s,d,m}^{disch,PEV}$ The bounded output decision variable (positive real number) regulates the charging and discharging of active power for the PEV
- $\delta_{i,t,s,d,m}^{ch,(SBESS,MBESS)}, \delta_{i,t,s,d,m}^{disch,(SBESS,MBESS)}$ The bounded output decision variable (positive actual number) regulates the SBESS and MBESS's charging and discharging active power
- M_1, M_2 Two positive variables
- $\gamma_{j,m}^{MBESS}$ The binary variable indicates whether the MBESS relocated compared to its location in the previous month
- $\lambda_{i,t,s,d,m}^{ch,PEV}, \lambda_{i,t,s,d,m}^{disch,PEV}$ The binary variables indicate the PEV's current operating status and prevent simultaneous charging and discharging
- $\lambda_{i,t,s,d,m}^{ch,BESS}, \lambda_{i,t,s,d,m}^{disch,BESS}, \lambda_{i,t,s,d,m}^{floating,BESS}$ The binary variables obtained from the SBESS/MBESS operating status will control the SBESS/MBESS cycle count
- $EV_{i,t,s,d,m}^j$ The binary variable indicates if the PEV is currently parked
- $\tau_i^W, \tau_i^{PV}, \tau_i^{(PEV-PL)}, \tau_i^{SBESS}, \tau_i^{MBESS}$ The binary variables indicate the current availability of wind DG, PV DG, PEV-PL, SBESS, and MBESS

Objective functions parameters and variables

- ψ The vector consists of three objective functions to be minimized simultaneously
- ψ_1, ψ_2, ψ_3 The three proposed objective functions
- $\psi_W^{Invest}, \psi_{PV}^{Invest}, \psi_{(PEV-PL)}^{Invest}, \psi_{SBESS}^{Invest}, \psi_{MBESS}^{Invest}$ The functions represent the total annual investment cost of the assigned wind DGs/PV DGs/PEV-PLs/SBESSs/MBESSs (\$/year)
- $\psi_W^{Man}, \psi_W^{Man}, \psi_{(PEV-PL)}^{Man}, \psi_{SBESS}^{Man}, \psi_{MBESS}^{Man}$ The functions represent the total annual maintenance cost of the assigned wind DGs/PV

DGs/PEV-PLs/SBESSs/MBESSs (\$/year)

- $\psi_{DSS}^{oper}, \psi_W^{oper}, \psi_{PV}^{oper}, \psi_{(PEV-PL)}^{oper}, \psi_{SBESS}^{oper}, \psi_{MBESS}^{oper}$ The functions represent the total annual expected operation cost of the assigned wind DGs/PV DGs/PEV-PLs/SBESSs/MBESSs (\$/year)
- ψ_{DSS}^{Em} The function represents the DSS's total expected emission cost (\$/year)
- $\psi_{(PEV-PL)}^{buy}, \psi_{(PEV-PL)}^{sell}$ The functions represent the annual expected energy expenses and revenues from buying/selling energy to parked PEVs (\$/year)
- ψ_{MBESS}^{Transp} The function represents the total annual cost based on the monthly transportation schedule for the MBESSs (\$/year)
- $f_W^{Invest}, f_{PV}^{Invest}$ The unit investment cost for each deployed wind DG and PV DG maximum capacity (\$/MVA)
- $f_{(PEV-PL)}^{EGmax}, f_{(PEV-PL)}^{Amax}$ The unit investment cost for the maximum energy (\$/MWh) and power capacities (\$/MVA) for each deployed PEV-PL
- $f_{SBESS/MBESS}^{EGmax}, f_{SBESS/MBESS}^{Amax}$ The unit investment cost for the maximum energy (\$/MWh) and power capacities (\$/MVA) for each deployed SBESS/MBESS
- $\omega_W, \omega_{PV}, \omega_{(PEV-PL)}, \omega_{SBESS}, \omega_{MBESS}$ The capital recovery factor converts the capital cost into annual cost
- IR, IF The interest and inflation rate
- $f_{t,s,d,m}^{EP}$ The energy prices derived from MCS and BRM (\$/MWh)
- f_{DSS}^{Em} The unit emission cost for the delivered power by the DSS (\$/MWh)
- $f_W^{oper}, f_{(PEV-PL)}^{oper}, f_{SBESS}^{oper}, f_{MBESS}^{oper}$ The unit operation cost for the current operating power of each deployed wind DGs/PV DGs/PEV-PLs/SBESSs/MBESSs (\$/MVA)
- $f_{(PEV-PL)}^{Tarrif-buy}, f_{(PEV-PL)}^{Tarrif-sell}$ The designated tariff for buying/selling transactions between PEV-PL and parked PEVs (\$/MWh).
- $f_{fix}^{trip}, f_{var}^{trip}$ The unit fixed (\$/trip) and variable (\$/mile/trip) transportation costs for each MBESS

Technical parameters and variables

- $A_{j,W}^{max}, A_{j,PV}^{max}, A_{j,(PEV-PL)}^{max}, A_{j,(SBESS,MBESS)}^{max}$ The maximum apparent power for each of the wind DGs, PV DGs, PEV-PLs, SBESSs, and MBESSs (MVA)
- $A_{i,Wu}^{max}, A_{i,PVU}^{max}, A_{i,PEV}^{max}, A_{i,Battu}^{max}$ The maximum apparent power for each of the wind turbine units, PV units, PEVs, and SBattu/MBattu (MVA)
- $A_{j,t,s,d,m}^{DSS}, A_{j,t,s,d,m}^{W}, A_{j,t,s,d,m}^{PV}, A_{j,t,s,d,m}^{(PEV-PL)}, A_{j,t,s,d,m}^{SBESS}, A_{j,t,s,d,m}^{MBESS}, A_{j,t,s,d,m}^{PEV}$ The hourly variable apparent power for each of the DSSs, wind DGs, PV DGs, PEV-PLs, SBESSs, MBESSs, and PEVs (MVA)
- $A_{ij,t,s,d,m}^{brch}, A_{ij,t,s,d,m}^{rated}$ The hourly variable and rated apparent power for each branch (MVA)
- $A_{ij,t,s,d,m}^{Loss}$ The hourly apparent power loss in a branch between bus i and bus j (MVA)
- $AP_{j,W}^{max}, AP_{j,PV}^{max}$ The maximum active power for each of the wind DGs and PV DGs (MW)
- $AP_{i,PEV}^{max}, AP_{i,PEV}^{rated}$ The maximum and rated active power for each PEV (MW)
- $AP_{j,BESS}^{max}, AP_{j,BESS}^{rated}$ The maximum and rated active power for each SBESS and MBESS (MW)
- $AP_{j,t,s,d,m}^W, AP_{j,t,s,d,m}^{PV}$ The hourly variable active power for each wind DG and PV DG (MW)
- $AP_{j,t,s,d,m}^{buy,(PEV-PL)}, AP_{j,t,s,d,m}^{sell,(PEV-PL)}$ The hourly active power variation for each PEV-PL during energy purchase/sale transactions with parked PEVs (MW)
- $AP_{i,t,s,d,m}^{ch,PEV}, AP_{i,t,s,d,m}^{disch,PEV}$ The hourly variations in charging and discharging active power for each PEV (MW)

$AP_{j,t,s,d,m}^{ch,BESS}, AP_{j,t,s,d,m}^{disch,BESS}$	The hourly variations in charging and discharging active power for each SBESS and MBESS (MW)
AP_{PEV}, AP_{BESS}	A specific active power value for the PEV and SBESS/MBESS (MW)
$AP_{i,t,s,d,m}^{DSS}$	The hourly variations in active power for each DSS (MW)
$AP_{j,DSS}^{min}, AP_{j,DSS}^{max}$	The minimum and maximum active power values for each DSS (MW)
$AP_{i,t,s,d,m}^{CL}, AP_{i,t,s,d,m}^{RL}$	The hourly variable active power for each of the commercial and residential loads without DRP (MW)
$AP_{i,t,s,d,m}^{injected}, AP_{i,t,s,d,m}^{calculated}$	The hourly variable active power is injected into/transmitted from bus i (MW)
$AP_{ij,t,s,d,m}^{Loss}$	The hourly variable active power along a branch connected between bus i and bus j (MW)
$RP_{j,t,s,d,m}^W, RP_{j,t,s,d,m}^{PV}, RP_{j,t,s,d,m}^{(PEV-PL)}$	The hourly variable reactive power for each wind DG, PV DG, and PEV-PL (MVAR)
$RP_{i,t,s,d,m}^{disch,PEV}, RP_{j,t,s,d,m}^{disch,BESS}$	The hourly variable discharged reactive power for each PEV and SBESS/MBESS (MVAR)
$RP_{j,BESS}^{max}$	The assigned maximum reactive power for each SBESS/MBESS (MVAR)
$RP_{j,DSS}^{min}, RP_{j,DSS}^{max}$	The assigned minimum and maximum reactive power for each DSS (MVAR)
$RP_{i,t,s,d,m}^{CL}, RP_{i,t,s,d,m}^{RL}$	The hourly variable reactive power for each of the commercial and residential loads without DRP (MVAR)
$RP_{i,t,s,d,m}^{injected}, RP_{i,t,s,d,m}^{calculated}$	The hourly variable reactive power is injected into/transmitted from bus i (MVAR)
$RP_{ij,t,s,d,m}^{Loss}$	The hourly variable reactive power along a branch connected between bus i and bus j (MVAR)
$EG_{j,(PEV-PL)}^{max}, EG_{j,t,s,m,y}^{(PEV-PL)}$	The maximum energy capacity and hourly variation assigned to each PEV-PL (MWh)
$EG_{i,s,d,m}^{arr,PEV}, EG_{i,s,d,m}^{dep,PEV}$	The assigned arrival and departure energy capacity allocated to each PEV (MWh)
$EG_{i,PEV}^{min}, EG_{i,PEV}^{rated}, EG_{i,PEV}^{max}, EG_{i,t,s,d,m}^{PEV}$	The assigned minimum, rated, maximum, and hourly variable energy capacity for each PEV (MWh)
$EG_{i,BESS}^{initial}, EG_{i,BESS}^{min}, EG_{i,BESS}^{rated}, EG_{i,BESS}^{max}, EG_{i,t,s,d,m}^{BESS}$	The assigned initial, minimum, rated, maximum, and hourly variable energy capacity for each SBESS/MBESS (MWh)
$EG_{i,Battu}^{max}$	The assigned maximum energy capacity for each internal battery unit (MWh) for the SBESS/MBESS
$SOC_{i,PEV}^{min}, SOC_{i,PEV}^{max}$	The assigned minimum and maximum state of charge for each PEV (MWh)
$SOC_{i,BESS}^{initial}, SOC_{i,BESS}^{min}, SOC_{i,BESS}^{max}$	The assigned initial, minimum, and maximum state of charge for each SBESS/MBESS (MWh)
$N_{j,t,s,d,m}^{PEV}$	The number of parked PEVs in j^{th} PEV-PL at the current hour
N_{cyc}^{BESS}	The designated number of charging/discharging cycles for the SBESS/MBESS is configured to one
Δt	The designated scheduling time horizon is set to one hour (h)
$\xi_{ch}^{PEV}, \xi_{disch}^{PEV}, \xi_{ch}^{BESS}, \xi_{disch}^{BESS}$	The assigned charging and discharging efficiency for the PEV and SBESS/MBESS.
$\alpha_j^W, \alpha_j^{PV}, \alpha_i^{PEV}, \alpha_j^{BESS}$	The assigned power factor for each wind DG, PV DG, PEV, and SBESS/MBESS
$Lf_{W/PV/(PEV-PL)/SBESS/MBESS}$	The corresponding lifetime for the wind DG, PV DG, PEV-PL, SBESS, and MBESS (years)
$V_{j,t,s,d,m}$	The hourly voltage is applied to bus j (pu)
$\varphi_{ij,t,s,d,m}$	The variation in the phase angle difference between bus i and bus j (radian)
V_j^{min}, V_j^{max}	The assigned minimum and maximum allowed voltage on bus j (pu)
$oneX_{j,(m,m-1)}^{Transp}$	The transportation distance between the assigned locations of the MBESS in the current and previous months (mile)
G_{ij}, B_{ij}	The conductance and susceptance of the branch connected between bus i and bus j (s)
Probabilistic distribution parameters	
$ws_{j,t,s,d,m}$	The wind speed was obtained using MCS and BRM
$ws_{ci}, ws_{co}, ws_{no}$	The cut-in, cut-out, and nominal wind speed
$sr_{j,t,s,d,m}$	The obtained solar irradiation using MCS and BRM
f_{TG}	The truncated normal PDF
$t_{i,s,d,m}^{arr,PEV}, t_{i,s,d,m}^{dep,PEV}$	The obtained arrival and departure time of PEV using MCS and BRM
$\mu_{i,time}^{arr,PEV}, \mu_{i,time}^{dep,PEV}$	The assigned mean value for the arrival and departure time of PEV
$\sigma_{i,time}^{arr,PEV}, \sigma_{i,time}^{dep,PEV}$	The assigned standard deviation value for the arrival and departure time of PEV
$t_{i,min}^{arr,PEV}, t_{i,min}^{dep,PEV}$	The assigned minimum arrival and departure time for PEV
$t_{i,max}^{arr,PEV}, t_{i,max}^{dep,PEV}$	The assigned maximum arrival and departure time for PEV
$SOC_{i,s,d,m}^{arr,PEV}, SOC_{i,s,d,m}^{dep,PEV}$	The obtained arrival and departure state of charge of PEV using MCS and BRM
$\mu_{i,SOC}^{arr,PEV}, \mu_{i,SOC}^{dep,PEV}$	The assigned mean value for the arrival and departure SOC of PEV
$\sigma_{i,SOC}^{arr,PEV}, \sigma_{i,SOC}^{dep,PEV}$	The assigned standard deviation value for the arrival and departure SOC of PEV
$SOC_{i,min}^{arr,PEV}, SOC_{i,min}^{dep,PEV}$	The assigned minimum arrival and departure SOC for PEV
$SOC_{i,max}^{arr,PEV}, SOC_{i,max}^{dep,PEV}$	The assigned maximum arrival and departure SOC for PEV

abundance and redistributing it to the grid during peak demand. However, integrating EVs introduces additional technical complexities associated with uncertain variables such as time, distance, and energy, potentially constraining the system's capacity to accommodate a large fleet of EVs smoothly [4–6].

Moreover, propelled by advancements in power electronics technology, incorporating energy storage systems (ESS) emerges as the premier solution to tackle the inherent unpredictability of RESs. ESS provides the critical capability to store excess energy during periods of heightened RES generation variability, releasing stored energy during phases of high demand. This dynamic functionality plays a pivotal role

in enhancing the overall integration efficiency of RESs, thereby facilitating the widespread adoption of PEVs within the system. However, strategic decisions regarding optimal placement, capacity, power rating, and charging-discharging profiles must be made to fully leverage the potential benefits of deploying storage devices in DS. This underscores the necessity for employing intelligent programming algorithms for the optimal configuration and management of ESS resources. [8–10]. Additionally, the current shift towards embracing Mobile Battery Energy Storage Systems (MBESSs) [11] represents a significant stride in bolstering the integration of RESs and EVs. MBESSs present the advantage of mobility, allowing for temporary relocation based on optimal

cost-effectiveness considerations, thereby adapting to dynamic weather patterns and fluctuations in energy demand. Their ability to swiftly move to disaster-prone areas enhances system resilience to an even greater extent and is helpful for mobile charging stations for EVs [12].

1.2. Aim

This paper presents a comprehensive strategy for optimizing joint stochastic planning, focusing on determining the ideal locations and capacities of various components, including wind DGs, PV DGs, PEV-PLs, SBESS, and MBESS. Furthermore, it aims to optimize DG operation, manage the energy expenses and profits of PEV-PLs, schedule the monthly operations of MBESS, and charge/discharge SBESS/MBESSs. Moreover, the proposed planning model employs a stochastic approach to address uncertainties arising from variables such as wind speed, solar irradiance, load power, energy prices, and the energy and time dynamics of PEVs. The primary aim is to substantially enhance the integration of green energy, encompassing RESs and PEVs. Such integrated optimal planning and operation ensure a reliable power supply to meet the demands of traditional loads, PEVs, and SBESS/MBESS while upholding system security and accounting for technical constraints.

1.3. Literature review

Numerous research endeavors have been undertaken to tackle the intricate challenges surrounding the effective integration of EVs into complex and hybrid power and energy systems. The power and energy systems literature has outlined several critical components and keywords pertinent to this integration. These encompass EVs, PEV-PL, Fast Charging Stations (FCS), V2G technology, Hydrogen Powered EVs (HPEV), hybrid electric vehicles (HEV), wind DG, PV DG, fixed BESS (FBESS), MBESS, Hydrogen Energy Storage Systems (HESS), Thermal ESS (TESS), compressed air ESS (CAES), bidirectional Converters (BDC), Combined Heat and Power (CHP) units, biomass waste unit (BW), Gas boilers, Fuel Cells, Diesel distributed generators (DDGs), Energy Hubs (EHs), Energy Communities (ECs), Integrated Hybrid Power and Heat Energy Systems (IHES), Aggregators, Virtual Power Plants (VPP), mechanisms for upward and downward market regulation, reserves, Energy management system (EMS), Smart Homes (SHs) and buildings, demand side management (DSM), demand response program (DRP), network reconfiguration (NR), resilience to contingent events, intelligent programming, Internet of thing (IoT), Uncertainties, Forecasting methodologies, standalone and grid connected Microgrid (MG), DS, Transmission Systems (TS), electric, thermal and hydrogen loads, net energy metering (NEM), independent operators, and electricity markets.

1.3.1. Optimal planning and operation considering MBESS and RESs

In [11], utilizing a MINLP planning model, optimizing wind DGs, PV DGs, MBESS, and FCSs was conducted to enhance green energy integration and facilitate smart charger deployment within the DS. In [13], a joint operation optimization was proposed that combines the newly introduced MBESS for integration with existing wind DG, PV DG, and hydropower in the DS. The problem was formulated as a MINLP based on deep reinforcement learning (DRL) to determine the optimal charging-discharging profiles of the MBESS. It was proven that the deployment of MBESS decreases the operational cost of the DS, alleviates congestion, and increases the penetration of RESs. In a separate study in [14], the authors proposed a stochastic service restoration model that integrates MBESS fleets, MGs reconfiguration, and RESs. Additionally, joint probabilistic constraints were utilized to handle RES uncertainty, where a novel computational method with discrete probability density function (PDF) scenarios was used for a fast solution. The effectiveness and scalability of the proposed framework were demonstrated on IEEE 33-bus and IEEE 123-bus test systems, enhancing the DS's resilience against high-impact, low-probability disaster events. A MILP optimization problem was formulated in [15] to determine the

newly relocated optimal location, the starting time of transportation, and charging-discharging scheduling for the grid-scale MBESS mounted on a truck or train. The MBESS surpassed the SBESS in minimizing the total DS operation and load-shedding costs. However, RESs or EVs were not considered. Furthermore, in [16], a novel three-level defender-attacker-defender resiliency-oriented planning and operation model was proposed for optimizing the sizing and positioning of MBESS in networked MMGs. It utilized an adaptive genetic algorithm for the planning level and a consensus algorithm for multi-stage decentralized operational control. This approach determined energy sharing among MMGs, considering uncertainties through stochastic programming and accounting for external contingencies. MBESS proved more effective for load restoration than SBESS during extreme events, with capacities increasing with larger attack budgets.

In [17], a Model Predictive Controller (MPC) was employed as the centralized EMS for a smart home powered by the grid and rooftop PV solar panels linked to its internal SBESS. Additionally, the homeowner possesses an EV, with the EV battery serving as an MBESS when plugged into the home. The MPC proposed a single objective function to efficiently control the smart home's electrical and heating loads, manage energy flow, and effectively regulate the charging and discharging of the SBESS and MBESS, all while prioritizing comfort and minimizing energy costs. Monte Carlo simulation (MCS) was adopted to handle the uncertainties related to the arrival and departure times, as well as the energy requirements, of the EV. The study in [18] considered the mobile storage capability of EVs. The work aimed to optimize MMG systems amidst growing EV presence by proposing a clustering algorithm to expedite scheduling efficiency. The clustering technique addressed EV arrival/departure challenges across MGs by maximizing controllable time within parking. Moreover, it considered EV charging cost, power curve optimization, and renewable energy absorption. Simulation validated enhanced MMG performance and computational efficiency with a clustering algorithm, especially for large-scale EV integration. A networked grid-connected MG solution was utilized in [19] to improve the system resilience of the DS during contingency events. Each MG encompasses loads; EVs act as MBESS, SBESS, PEV-PL, DDGs, and wind DG. The optimal coordination between the proposed local and central EMSs was designed as a multi-level problem where the mobile battery of the EVs parked in the PEV-PL of the grid-connected MGs was used to supply power to the loads existing in the islanded MGs due to an outage. Moreover, authors in [20] proposed a new model for managing MBESSs as mobile charging stations (MBESS-CS) in the DS with fixed charging stations, wind DG, and PV DG. The MBESS-CS utilized truck-mounted BESS with EV charging sockets where the DS network charges the MBESS-CS during low-demand periods. Then, the stored energy in the MBESS-CS was used to charge EVs accumulated beside an optimally determined fixed charging station location. Moreover, the model optimized the spatial-temporal schedule and internal battery's charging/discharging of the MBESS-CS. It was proved that MBESS-CS reduces EV queue at fixed stations, meets charging demand using off-peak energy, and reduces electricity costs for charging.

1.3.2. Optimal planning and operation considering PEVs, RESs, and SBESS

In [21], the authors introduced a strategic decade-long planning model incorporating wind DGs, PV DGs, and SBESSs. The primary focus was improving green energy integration by optimally determining the selected sources' location, capacity, and operation to minimize three key objectives: the total average planning and operational costs over the project's lifespan, bus voltage fluctuations, and feeder power losses. This was further expanded in [22] to incorporate DRP. Furthermore, the study in [22] was restructured into a two-stage optimization framework in [23], integrating a NR procedure. Moreover, The study in [24] presented a three-level optimization model for sizing networked MGs, balancing resilience and cost. It uses Adaptive Genetic Algorithm (AGA) for sizing and attack generation, NSGA-II for multi-objective optimization, and time-coupled AC OPF for accuracy. Internal uncertainties and

external contingencies are integrated using a defender-attacker-defender approach. Case studies show networked MGs reduce costs while enhancing resilience compared to standalone systems. Key components include back-up generators, SBESS, and critical vs. non-critical loads. The model identifies trade-offs, demonstrating that higher resilience requires more investment but improves load-shedding performance. However, it is noteworthy that the authors in [21–24], did not include PEV-PL or MBESS within their developed model. The authors in [25] built upon the methodology presented in [21] by incorporating the optimal integration of PEV-PL. This work primarily focuses on highlighting the significance of deploying PEV-PL in enhancing the penetration levels of EVs and the capacity of integrated RESs, particularly wind and PV systems. However, the methodology omitted the role of SBESSs, paving the way for EVs to serve as mobile storage devices.

In [26] an optimal planning model for short-term objectives was presented, aiming to determine the optimal placement, capacity, and operation scheduling of wind farms, PV farms, SBESSs, and PEV-PLs while incorporating DRP to bolster resilience within the DS. In another study in [27], a planning model was introduced to optimize the operation of DS by strategically locating RESs-DGs, SBESS, and EV charging stations. The model accounted for factors such as battery degradation and uncertainties arising from wind, solar generation, load, and EV charging pattern fluctuations. Additionally, authors in [28] introduced a sizing and techno-economic assessment issue tackled with the Modified Salp Swarm Algorithm (MSSA). The goal was to ascertain the optimal capacities of an FCS, wind DG, PV DG, SBESS, and related BDC within grid-connected setups across three varied Indian environmental contexts. A novel planning challenge was formulated in [29] to pinpoint the best placement, capacity, and power rating of CAESs and the most effective operation strategy for Ts. This task integrated multiple correlated uncertainties such as wind speed, solar irradiation, load power, and energy prices.

Recent research has delved into IHES, incorporating EVs. Authors in [30] introduced a short-term optimization challenge for a hybrid standalone utterly renewable energy system (SHARES) comprising wind DG, PV DG, BW units, TESS, HESSs, PEV-PLs, and associated –BDCs. The primary aim was to efficiently supply energy at minimum cost for electrical and thermal requirements in this hybrid electric and heating HSARES. An IHES tailored for a network of interconnected residential buildings has also been extensively investigated [31]. The studied system incorporated gas boilers, CHP-MT, rooftop solar PV panels, SBESS, BDC, ten private EV FCSs, 20 shared EV FCSs, and NEM. Moreover, incentive-driven DRPs were leveraged to optimize operational efficiency within the system. Moreover, a novel approach was presented in [32] for the long-term planning of PEV-PL with a lifespan of 10 years. This method integrated probability analysis for EV adoption, optimal scheduling of charging/discharging, and modeling of EV proliferation. It facilitated optimal decision-making regarding the location, capacity, and timing of PEV-PL expansion. In [33], two ownership models for FCS incorporating RESs were proposed, focusing on FCS and power grid perspectives. Wind power was forecasted using a fuzzy neural network and an improved PSO algorithm to reduce uncertainties. MILP model optimized via GAMS to maximize 10-year project net present cost. DSM and DRPs lower costs by 17.85 % and 3.31 % in respective ownerships. A 25-year project was initiated in [34] for Grid-Tied Solar PV Highway Charging Stations, aiming for efficient scheduling to accommodate 250 EVs daily. The optimization targeted cost reduction covering initial investment, operational expenses, energy revenue, and PV converter replacement. Furthermore, authors in [35] detailed a project at Shah Amanat International Airport in Bangladesh for sizing and economic analysis of a grid-connected PEV-PL. It integrated wind and PV systems, employing Fuzzy logic and Recurrent Neural Networks-Long Short-Term Memory (RNN-LSTM) for load forecasting. The objective was to minimize costs and maximize profits through optimal sizing and operation.

To assess the impact of EVs on system resilience, [36] introduced a bi-level model to enhance system resiliency by optimizing islanding

decisions during network faults. Considering economic incentives, it prioritized load recovery via PEV charge/discharge management. The upper level focused on DS islanding for load restoration, while the lower level determined energized PEV-PLs allocation. Additionally, [37] it has introduced a novel method for bolstering building energy resilience during disasters. It combined Chicken Search Optimization and Spike Neural Network learning, incorporating shared EV parking stations. The approach utilized power management strategies like peer-to-peer building operations and DRP for cost reduction and resilience improvement without extra components. Authors in [38] analyzed an advanced power system for enhanced resilience in smart DS, minimizing daily operating and energy not supplied (ENS) costs. Components include intelligent PEV-PLs, RESs (wind, PV DGs), DDGs, SBESS, and DRP-capable intelligent consumers. Considering other contingency events, in [39], the authors introduced a graph-based approach for EV impact assessment on DSs and SESS economic viability. Analysis of 6500 DSs showed increased overloads with EVs, mostly near transformers. ESS became more economically feasible with lower storage costs, offering substantial cost savings with higher EV penetration. Moreover, [40] used BD-PEVs to enhance grid frequency, analyzing total harmonic distortion (THD) via bi-directional chargers. The aim was to maximize BD-PEV capacity for frequency regulation during grid frequency decline due to unforeseen events.

The study in [41] proposed data-driven uncertainty sets and an enhanced hierarchical framework for efficient EM in MMG systems. It addressed challenges posed by RES variability and EV integration, aiming to overcome existing method limitations. Additionally, the work in [42] suggested that high-control HEVs boost grid flexibility and address energy and environmental concerns. The study proposed an EMS employing bald eagle search optimizer for MGs with RESs and HEVs to minimize costs and emissions, accounting for various RES scenarios and EV charging modes. Moreover, [43] proposed an optimal scheduling solution for a MG using a modified fluid search optimization algorithm (MFSO). The MG includes wind, PV, fuel cells, and two MTs, with two V2G PEV fleets totaling 68 and 40 vehicles. Effective PEV fleet routing reduced uncertainties, while V2G PEV integration cut MG operational costs. Furthermore, [44] targeted minimizing operating costs and PEV charging expenses in a MG using PV power. It employed an online MILP control strategy with an adaptive rolling horizon (ARH) to optimize all PEVs. The study discussed in [45] investigated the optimal scheduling of a nano-grid (NG) comprising a mobile base station, rooftop solar PV panel, diesel generator, stationary HESS, and three PEVs. The problem was formulated as a mixed-integer second-order cone programming (MISOCP) to capture nonlinearities for optimization to minimize a single objective function, including both emission and operation costs.

1.3.3. Optimal scheduling considering PEVs, RESs, SBESS, and MBESS

The study in [46] introduces a multi-agent deep reinforcement learning (MADRL) framework using parameterized double deep Q-networks to optimize the routing and scheduling of MBESSs for resilience enhancement. By reformulating the problem as a Partially Observable Markov Game, it integrates power and transportation networks while addressing uncertainties in renewable energy, demand, line outages, and traffic. The method employs linearized AC-OPF and dynamic NR for technical accuracy. Case studies on 6-bus and 33-bus systems show its efficiency and adaptability, outperforming model-based approaches and enabling scalable, decentralized decision-making for resilient energy systems. Another scheduling problem for MBESSs in MGs was investigated in [47]. This study proposes a framework for resilient load restoration by coordinating MBESSs and stationary resources using a Markov Decision Process solved with Twin Delayed Deep Deterministic Policy Gradient. The approach integrates dynamic MESS routing within a transportation network, addressing uncertainties in load demand and renewable energy. It optimizes load restoration while minimizing costs and meeting system constraints. Case studies with three MGs and a

transportation network show improved resilience, efficiency, and adaptability in managing hybrid energy systems under uncertainty.

Authors in [48] introduced a two-level MILP optimization framework, incorporating day-ahead scheduling and real-time operation, to optimize PEV-PL operator engagement in energy, reserve, and market mechanisms within the DS. It considered wind and PV DG locations and sizes, aiming to minimize DS operation costs and maximize operator benefits. The study in [49] developed a 24-hour EM strategy for five classes of grid-connected PEVs, integrating wind and PV systems. The Reinforcement Learning (RL) model based on the Deep Q Network (DQN) algorithm handled uncertainties from EV parameters, wind, and solar. Additionally, [50] presented a scheduling tool featuring Intelligent PEV-PL for ECs. It utilized a two-stage approach with a hybrid stochastic IGDT model to manage uncertainties. Simulations validated economic benefits but cautioned against overloading to avoid increased operating costs. Moreover, [51] explored interactions between intelligent PEV-PL, HESS, RESs (wind, PV), and the grid, integrating 15 G2V-capable HEVs. Optimal charging-discharging schedules were coordinated, addressing uncertainties via the Two-Point Estimation Method (2-PEM) for variables like wind speed, solar radiation, temperature, and PEV-PL load.

The work in [52] presented the Technical VPP (TVPP) model integrating DERs, PEV-PL, and DRPs for complex buildings. Unlike traditional VPPs, it optimized economically and technically feasible schedules, considering network constraints. The study examined three TVPPs enabling energy trading within and between TVPPs and the external grid, enhancing DERs' schedules, system reliability, and DRP engagement, thus reducing costs and imported power. Authors in [53] developed a comprehensive scheduling framework integrating DS and aggregator operators, various DERs, and PEVs, considering normal and interrupted modes. The model comprised a two-stage, four-level scheduling for day-ahead charging, DER, aggregator, and DS operators, replicated for real-time market scheduling. Additionally, [54] introduced novel model-free multi-agent deep reinforcement learning (MADRL) to optimize coordination in interconnected multi-energy systems with EH and PEV aggregators. It aimed to minimize EH operation costs and maximize aggregator profits amid uncertainties. Moreover, [55] formulated a two-stage optimization for day-ahead and real-time markets in a smart DS. It includes PV DGs, DDGs, HPV fueling stations, SESSs, and a time-of-use (ToU) DRP. Uncertainties were addressed using a scenario-based approach for HPV behavior, solar irradiation, market prices, and energy/hydrogen demands. Furthermore, [56] introduced a three-stage energy, reserve, and market regulation management framework across interconnected TS and DS. SHs engaged in daily planning using IoT-based appliances and EVs, the DS operator devises market strategies, and the TS operator settles markets. MILP optimized via GAMS highlights SHs' control benefits, with large-scale industrial DRP engagement reducing costs and improving voltage profiles. Authors in [57] introduced an AI-powered EV charging hub with 150 stations and SBESSs. It optimized power usage by committing most needs in the day-ahead market and using SBESS with real-time market purchases. MILP handles day-ahead commitments, while deep reinforcement learning (DRL) manages real-time operations.

1.4. Research gaps and contributions

Drawing upon an exhaustive literature review and harnessing the authors' collective expertise, it becomes apparent that prior research has only scratched the surface regarding integrating MBESS into the DS. Some studies have explored the use of PEVs as a MBESS. Additionally, specific works have suggested hourly relocation of MBESS, a concept impractical in real-world scenarios. Furthermore, a scant few research endeavours have explored the combined integration of SBESS and MBESS to bolster RESs and PEVs integration. Significantly, the study in [11] represents the sole literature focusing on the practical monthly relocation of MBESS to augment the penetration level of RESs and FCSS.

However, several shortcomings were identified within this study:

- It is impractical to focus solely on MBESS without integrating SBESS. Since all MBESS units are relocated at the end of each month, the absence of SBESS units during the transportation process means there would be no storage devices available to fulfil their functions in the interim.
- Emphasis solely on economic objectives while disregarding technical considerations, which undoubtedly influence optimal cost outcomes.
- Limited analysis restricted to comparing system performance before and after MBESS allocation, neglecting the necessity to assess various system configurations.
- The concept of PEV-PL was not incorporated.

This paper addresses these limitations by introducing a comprehensive stochastic planning model. This model accounts for economic and technical objectives and facilitates the seamless integration of RESs (wind and PV), PEV-PL, SBESS, and MBESS within the DS. The aim is to bolster the capacity of deployed RESs and PEVs, thereby advancing decarbonization initiatives within the power and automotive sectors.

The primary contribution of this research can be encapsulated as follows:

- Constructing a stochastic MINLP optimization cooperative planning model that includes wind DGs, PV DGs, PEV-PLs, SBESS, and MBESSs to increase green energy integration in the DS.
- This planning model aims to achieve the optimal location, sizing, and scheduling of wind DGs, PV DGs, PEV-PLs, SBESSs, and MBESSs by simultaneously minimizing three objective functions: the annual total expected planning and operational cost, power loss, and voltage deviation.
- Developing a flexible and coordinated optimal charging and discharging control strategy for the PEVs, SBESS, and MBESSs. Furthermore, adopting a monthly transportation schedule for the MBESSs adds a practical dimension to the research.
- Various configurations were examined and juxtaposed to demonstrate the efficacy of the proposed planning model under typical operational modes. The findings from these configurations underscore the significance of incorporating a hybrid MBESS and SBESS to maximize green energy integration in both the RESs and transportation sectors.

1.5. Article structure

The remainder of this paper is as follows: Section 2 describes the proposed planning model framework. Section 3 formulates the proposed objective functions and the subjected constraints. Section 4 describes the test system under study, input profiles, and the studied cases and discusses the simulation results. The conclusions are summarized in section 5. Section 6 outlines this study's limitations and presents plans to address the identified gaps.

2. Model description

Integrating RDGs, particularly wind and PV sources, into radial electric power DSs introduces significant uncertainties. These stem from the intermittent and unpredictable nature of RDGs, presenting various technical and economic challenges for DS operators. These challenges encompass power quality issues, reliability concerns, and power curtailment.

ESS technology is pivotal in enabling high penetration levels of RDGs into DSs. By charging during periods of low demand and discharging during peak demand or high electricity price periods, ESSs facilitate optimal utilization of RDGs. However, maximizing the benefits of this integration requires DS operators to deploy ESSs strategically. This involves determining the optimal number, size, type, location, and

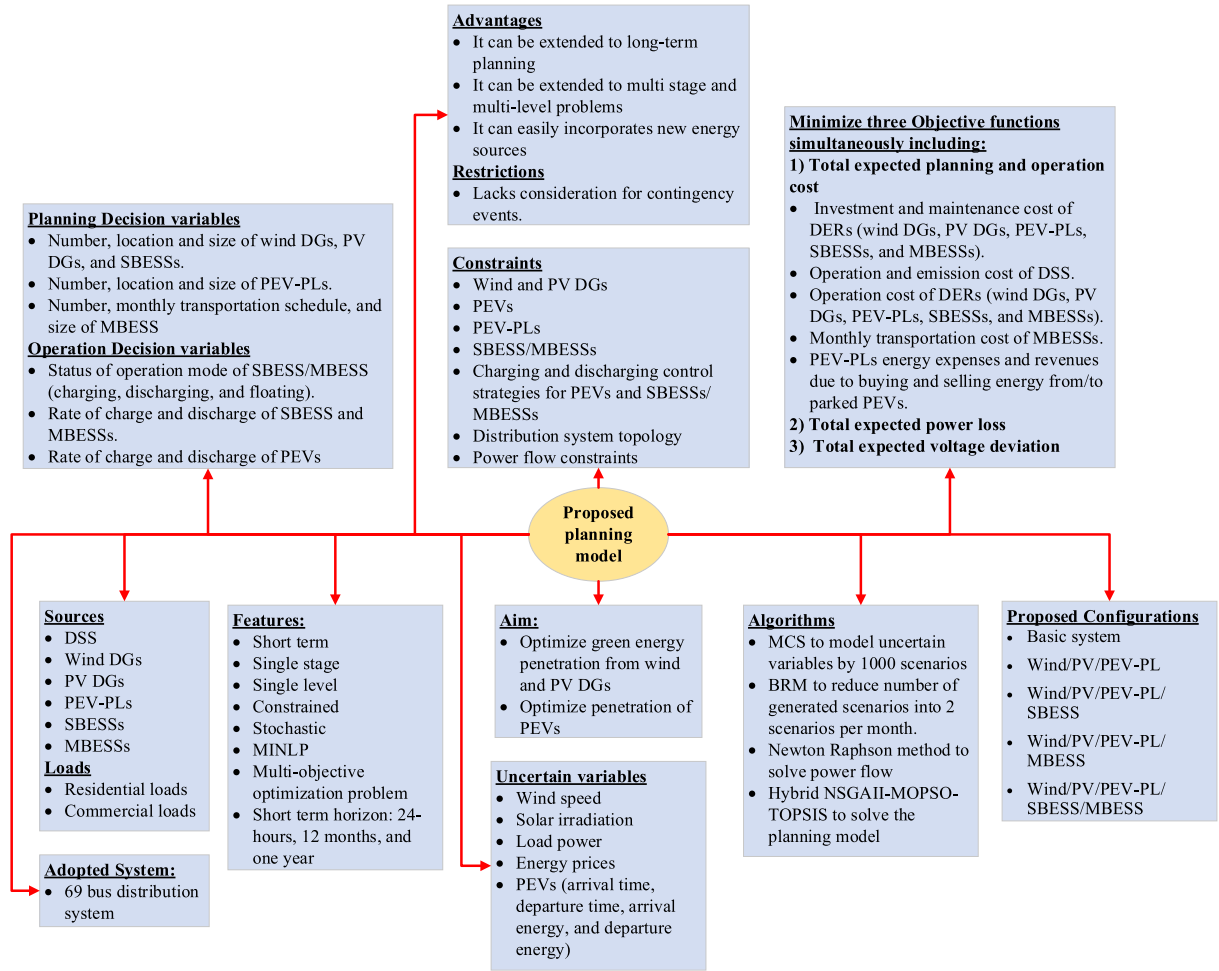


Fig. 1. The framework of the proposed planning model.

charging/discharging profiles of ESSs. Furthermore, introducing MBESSs adds another dimension to enhancing green energy penetration. MBESSs possess the flexibility to relocate monthly, adapting to variable weather conditions and load variations. Moreover, their ability to relocate to vulnerable areas post-disaster enhances system resilience further.

The substantial number of bidirectional PEVs connected to DSs can serve as additional mobile storage in addition to SBESSs and MBESSs. Coordinated operation with SBESSs and MBESSs can significantly increase RDG penetration levels, yielding additional benefits. Therefore, deploying PEV-PL is essential to facilitate interaction between PEVs and DSs through intelligent charging and discharging interfaces. DS operators actively seek comprehensive cooperation among RDGs, PEV-PLs, SBESSs, and MBESSs. This cooperation aims to optimize green energy integration and meet demand requirements.

The objective of this collaborative approach, incorporating wind DGs, PV DGs, PEV-PLs, SBESSs, and MBESSs, presumed to be under the ownership and operation of the DS, is to advance the decarbonization of both the power and automotive sectors by elevating the penetration level of RESs and PEVs. The framework of this comprehensive model is illustrated in Fig. 1, encompassing the following considerations:

- The radial DS topology and associated data, encompassing bus configurations, feeders, loads, and substations.
- Key attributes of this planning approach include its short-term horizon, single-stage structure, single-level complexity, stochastic nature, MINLP framework, and multi-objective constrained optimization objectives.
- Decision variables within the planning framework include:

- Determining the number, placement, and capacity of wind DGs, PV DGs, PEV-PLs, SBESSs, and MBESSs, as shown in (1)-(3).

$$\delta_{(W/PV/(PEV-PL)/SBESS/MBESS)}^{number} \in [1, N_{(W/PV/(PEV-PL)/SBESS/MBESS)}^{max}] \quad (1)$$

$$\delta_{j,(W/PV/(PEV-PL)/SBESS)}^{Location} \in [2, N_{bus}] \forall j \in \delta_{(W/PV/(PEV-PL)/SBESS)}^{number} \quad (2)$$

$$\delta_{j,(W/PV/(PEV-PL)/SBESS/MBESS)}^{size} \in [1, N_{Wu/PVu/PEVs/SBattu/MBattu}^{max}] \forall j \in \delta_{(W/PV/(PEV-PL)/SBESS/MBESS)}^{number} \quad (3)$$

- Establishment of a monthly transportation schedule for MBESSs, as in (4).

$$\delta_{j,m,MBESS}^{Location} \in [2, N_{bus}], \forall j \in \delta_{MBESS}^{number}, \forall m \in \Phi_M \quad (4)$$

- Operational decision variables consist of:
 - Status of SBESSs/MBESSs, encompassing charging, discharging, and floating modes, as in (5), where -1 , 0 , and $+1$ represent discharging, floating, and charging modes.

$$(\forall s \in \Phi_S, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M)$$

$$\delta_{j,t,s,d,m}^{status,(SBESS,MBESS)} \in \{-1, 0, +1\}, \forall j \in \delta_{SBESS/MBESS}^{number} \quad (5)$$

- Charge and discharge rates for PEVs, SBESSs, and MBESSs, as presented in (6) and (7).

$$(\forall s \in \Phi_S, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M)$$

$$0 \leq \left(\delta_{i,t,s,d,m}^{ch,PEV}, \delta_{i,t,s,d,m}^{disch,PEV} \right) \leq M_1, \forall i \in \delta_{j,(PEV-PL)}^{size}, \forall j \in \delta_{(PEV-PL)}^{number} \quad (6)$$

$$0 \leq \left(\delta_{j,t,s,d,m}^{ch,(SBESS,MBESS)}, \delta_{j,t,s,d,m}^{disch,(SBESS,MBESS)} \right) \leq M_2, \forall j \in \delta_{SBESS/MBESS}^{number} \quad (7)$$

- Consideration of uncertainties associated with variables such as wind power, solar power, load power, energy prices, and PEV's parking time and energy can be addressed by:
 - Extracting historical data on wind speed, solar radiation, load power, energy prices, and the arrival and departure times and energies of PEV drivers at the PEV-PLs.
 - Modeling the stochastic nature of these variables through suitable PDFs, subsequently sampled into 1000 scenarios using the presented MCS in [3] before being reduced to 2 scenarios per month using the BRM [58].
 - Utilization of Weibull and Beta PDFs to fit historical records of wind speed and solar irradiation while adopting the Normal PDF for fitting historical records of load power and energy prices.
 - Employing Truncated Gaussian distribution to model historical data related to the arrival time, departure time, arrival energy, and departure energy of PEVs.
- The optimization objectives encompass three functions to be minimized simultaneously:
 - Total annual expected costs associated with investment, maintenance, and operation of all considered planned and hourly operating decision variables, including wind DGs, PV DGs, PEV-PLs, SBESS, and MBESSs.
 - Total annual expected voltage deviation.
 - Total annual expected power loss.
- Constraints encompass DS upper-level substation (DSS), Wind DGs, PV DGs, PEVs, PEV-PLs, SBESSs, MBESSs, and power flow considerations.
- Implement a charging and discharging control strategy for SBESS, MBESSs, and PEVs.
- The solving methodology adopts a hybrid NSGAII-MOPSO-TOPSIS approach.

3. Problem formulation

This section presents the mathematical formulation of the objective functions and the subjected constraints in the proposed planning and collaborative model.

3.1. Objective functions

The proposed model effectively addresses three simultaneous objectives, as illustrated in (8). The primary objective function delineated in (9) encompasses the investment, maintenance, and operational costs associated with RDGs such as wind and PV systems, PEV-PLs, SBESSs, and MBESSs. The remaining objectives, articulated in (10) and (11), are technical, encompassing the total annual expected power loss and voltage deviation, respectively.

$$\psi = \text{Min}[\psi_1 \quad \psi_2 \quad \psi_3] \quad (8)$$

$$\psi_1 = \left\{ \begin{array}{l} \psi_W^{\text{Invest}} + \psi_{PV}^{\text{Invest}} + \psi_{(PEV-PL)}^{\text{Invest}} + \psi_{SBESS}^{\text{Invest}} + \psi_{MBESS}^{\text{Invest}} + \\ \psi_W^{\text{Man}} + \psi_{PV}^{\text{Man}} + \psi_{(PEV-PL)}^{\text{Man}} + \psi_{SBESS}^{\text{Man}} + \psi_{MBESS}^{\text{Man}} \\ + \psi_{DSS}^{\text{oper}} + \psi_{DSS}^{\text{Em}} + \psi_W^{\text{oper}} + \psi_{PV}^{\text{oper}} + \psi_{(PEV-PL)}^{\text{oper}} + \psi_{(PEV-PL)}^{\text{buy}} \\ - \psi_{(PEV-PL)}^{\text{sell}} + \psi_{SBESS}^{\text{oper}} + \psi_{MBESS}^{\text{oper}} + \psi_{MBESS}^{\text{Transp}} \end{array} \right\} \quad (9)$$

$$\psi_2 = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{bus}, i \neq j} |A_{ij,t,s,d,m}^{\text{Loss}}| \quad (10)$$

$$\psi_3 = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{bus}} |V_{j,t,s,d,m} - 1| \quad (11)$$

The total annual investment costs associated with wind DGs and PV DGs are contingent upon several factors, including the capital recovery factor, the maximum achievable optimal apparent power capacity (MVA), and the unit investment cost due to apparent power (\$/MVA), outlined in (12) and (13). Conversely, the total annual investment costs linked to deployed PEV-PLs, SBESSs, and MBESSs hinge upon the capital recovery factor, the optimal energy (MWh), and apparent power (MVA) capacities obtained, as well as the unit investment cost due to the energy (\$/MWh) and apparent power (\$/MVA), as depicted in (14)–(16). The capital recovery factor for each allocated component converts their capital costs into annual costs, determined by their respective lifetimes and the specified effective interest rate, as detailed in (17). Furthermore, the effective interest rate can be ascertained based on the defined interest and inflation rates, elucidated in (18). Furthermore, the annual maintenance cost for each allocated component is calculated as 2 % of the determined investment cost for each element, as indicated in (19).

$$\psi_W^{\text{Invest}} = \sum_{j \in \Phi_W} \varpi_W \times A_{j,W}^{\text{max}} \times f_W^{\text{Invest}} \quad (12)$$

$$\psi_{PV}^{\text{Invest}} = \sum_{j \in \Phi_{PV}} \varpi_{PV} \times A_{j,PV}^{\text{max}} \times f_{PV}^{\text{Invest}} \quad (13)$$

$$\psi_{(PEV-PL)}^{\text{Invest}} = \sum_{j \in \Phi_{PEV-PL}} \varpi_{(PEV-PL)} \times \left[EG_{j,(PEV-PL)}^{\text{max}} \times f_{(PEV-PL)}^{\text{Em}} + A_{j,(PEV-PL)}^{\text{max}} \times f_{(PEV-PL)}^{\text{Amax}} \right] \quad (14)$$

$$\psi_{SBESS}^{\text{Invest}} = \sum_{j \in \Phi_{SBESS}} \varpi_{SBESS} \times \left[EG_{j,SBESS}^{\text{max}} \times f_{SBESS}^{\text{Em}} + A_{j,SBESS}^{\text{max}} \times f_{SBESS}^{\text{Amax}} \right] \quad (15)$$

$$\psi_{MBESS}^{\text{Invest}} = \sum_{j \in \Phi_{MBESS}} \varpi_{MBESS} \times \left[EG_{j,MBESS}^{\text{max}} \times f_{MBESS}^{\text{Em}} + A_{j,MBESS}^{\text{max}} \times f_{MBESS}^{\text{Amax}} \right] \quad (16)$$

$$\varpi_{W/PV/(PEV-PL)/SBESS/MBESS} = \frac{IR'(1 + IR')^{L_{fW/PV/(PEV-PL)/SBESS/MBESS}}}{\left((1 + IR')^{L_{fW/PV/(PEV-PL)/SBESS/MBESS}} - 1 \right)} \quad (17)$$

$$IR' = \frac{IR - IF}{1 + IF} \quad (18)$$

$$\psi_{(W/PV/(PEV-PL)/SBESS/MBESS)}^{\text{Man}} = (2/100) \times \psi_{(W/PV/(PEV-PL)/SBESS/MBESS)}^{\text{Invest}} \quad (19)$$

The DSS's total expected cost encompasses operation and emission expenses. The DSS's total expected operational cost can be computed by leveraging the calculated joint probability of uncertain variables, the hourly supplied apparent power, and the energy prices outlined in (20). Similarly, the emission cost is contingent upon the same operational factors, albeit necessitating the utilization of the unit emission cost instead of the unit operational cost, as demonstrated in (21). Additionally, the annual expected operational cost of the wind DGs and PV DGs can be computed using the calculated joint probability of uncertain variables, the hourly supplied apparent power (MVA), and the unit cost for successful operations (\$/MVA) as illustrated in (22) and (23).

$$\psi_{DSS}^{\text{oper}} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{DSS}} \left(A_{j,t,s,d,m}^{\text{DSS}} \times f_{t,s,d,m}^{\text{EP}} \right) \quad (20)$$

$$\psi_{DSS}^{\text{Em}} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{DSS}} \left(A_{j,t,s,d,m}^{\text{DSS}} \times f_{DSS}^{\text{Em}} \right) \quad (21)$$

$$\psi_W^{\text{oper}} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_W} \left(A_{j,t,s,d,m}^W \times f_W^{\text{oper}} \right) \quad (22)$$

$$\psi_{PV}^{oper} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{PV}} \left(A_{j,t,s,d,m}^{PV} \times f_{PV}^{oper} \right) \quad (23)$$

Moreover, the total annual expected operational cost of the utilized PEV-PLs can be determined through the utilization of the calculated joint probability of uncertain variables, the magnitude of the hourly apparent power (MVA), and the unit cost for successful operations (\$/MVA), as depicted in (24). Furthermore, the deployed PEV-PLs can procure energy from the parked PEVs in discharging mode or vend energy to the parked PEVs in charging mode. Therefore, the total annual expected energy expenses/profits can be calculated in (25) and (26) by considering the joint probability of uncertain variables, along with the buying/selling of energy by the PEV-PLs from/to the parked PEVs in discharging/charging mode, and the predetermined tariff associated with these transactions.

$$\psi_{(PEV-PL)}^{oper} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{(PEV-PL)}} \left(A_{j,t,s,d,m}^{(PEV-PL)} \times f_{(PEV-PL)}^{oper} \right) \quad (24)$$

$$\psi_{(PEV-PL)}^{buy} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{(PEV-PL)}} \left(A_{j,t,s,d,m}^{buy,(PEV-PL)} \times f_{(PEV-PL)}^{Tariff-buy} \right) \quad (25)$$

$$\psi_{(PEV-PL)}^{sell} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{(PEV-PL)}} \left(A_{j,t,s,d,m}^{sell,(PEV-PL)} \times f_{(PEV-PL)}^{Tariff-sell} \right) \quad (26)$$

Furthermore, the total annual expected operational cost of the employed SBESSs and MBESSs can be determined by utilizing the calculated joint probability of uncertain variables, along with the magnitude of the hourly charging/discharging apparent power (MVA), and the unit cost for successful operations (\$/MVA), as outlined in (27)-(28). Additionally, the total annual expected cost of the allocated MBESSs encompasses operational expenses and the monthly scheduled transportation costs. The total yearly cost of the monthly-based transportation schedule for the MBESSs can be computed in (29), contingent upon a binary factor defined in (30), the predetermined fixed cost, and the variable transportation costs per unit distance, and the distance between the specified locations of the MBESS in the current and previous months. The distance X can be computed as shown in (31), where dx and dy represent the coordinates of the current and previous monthly bus locations of the MBESS.

$$\psi_{SBESS}^{oper} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{SBESS}} \left(A_{j,t,s,d,m}^{SBESS} \times f_{SBESS}^{oper} \right) \quad (27)$$

$$\psi_{MBESS}^{oper} = \sum_{m \in \Phi_M} \sum_{d \in \Phi_D} \sum_{s \in \Phi_S} \mathcal{E}_{s,d,m} \sum_{t \in \Phi_T} \sum_{j \in \Phi_{MBESS}} \left(A_{j,t,s,d,m}^{MBESS} \times f_{MBESS}^{oper} \right) \quad (28)$$

$$\psi_{MBESS}^{Transp} = \sum_{m \in \Phi_M} \sum_{j \in \Phi_{MBESS}} \gamma_{j,m}^{MBESS} \times \left(f_{fix}^{trip} + f_{var}^{trip} \times X_{j,(m,m-1)}^{Transp} \right) \quad (29)$$

$$\gamma_{j,m}^{MBESS} = \begin{cases} 1 & X_{j,(m,m-1)}^{Transp} \neq 0 \text{ \& } \delta_{j,MBESS}^{size} \neq 0 \\ 0 & \text{Otherwise} \end{cases} \quad (30)$$

$$X_{j,(m,m-1)}^{Transp} = |(dx_{j,m} - dx_{j,m-1})| + |(dy_{j,m} - dy_{j,m-1})| \quad (31)$$

3.2. Wind and PV DG constraints

This subsection mathematically formulates the constraints applied to wind DGs and PV DGs. The maximum apparent power, utilized to calculate the investment cost for each wind and PV DG, can be determined using (32) and (33), based on the maximum apparent power of internal wind turbines and PV units and the obtained optimal size. Additionally, the maximum active power for each wind DG and PV DG can be computed, as outlined in (34) and (35), leveraging the obtained optimal capacity and the designated power factor. Moreover, the hourly active power for each wind DG can be determined according to (36)

[29]. This computation relies on the obtained optimal capacity from (34), the wind speed, ws obtained from the MCS and BRM based on historical records and wind speed parameters such as cut-in, cut-out, and nominal values. Conversely, the computed hourly active power for each PV DG delineated in (37) [29] depends on the obtained optimal capacity from (35), the solar irradiation sr obtained from the MCS and BRM based on historical records, a designated radiation point sr_C set to 150 W/m^2 , and the solar radiation under standard conditions sr_{std} set to 1000 W/m^2 . Moreover, the hourly apparent power for each wind DG and PV DG, utilized for operational cost computation, can be determined based on the computed hourly active power from (36) and (37), along with the assigned power factor, as illustrated in (38) and (39). Additionally, the hourly delivered reactive power by the wind DG and PV DG can be calculated as demonstrated in (40) and (41), relying on the output variables computed in (38) and (39), as well as the assigned power factor.

$$\left(\forall j \in \Phi_{W/PV} = [1, \delta_{(W/PV)}^{number}], \forall i \in \Phi_{WU/PVU}^j = [1, \delta_{j,(W/PV)}^{size}], \forall s \in \Phi_S, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M \right)$$

$$A_{j,W}^{\max} = \delta_{j,W}^{size} \times A_{i,WU}^{\max} \quad (32)$$

$$A_{j,PV}^{\max} = \delta_{j,PV}^{size} \times A_{i,PVU}^{\max} \quad (33)$$

$$AP_{j,W}^{\max} = A_{j,W}^{\max} \times \alpha_j^W \quad (34)$$

$$AP_{j,PV}^{\max} = A_{j,PV}^{\max} \times \alpha_j^{PV} \quad (35)$$

$$AP_{j,t,s,d,m}^W = \begin{cases} 0 & \text{if } ws_{j,t,s,d,m} \leq ws_{ci} \text{ or } ws_{j,t,s,d,m} > ws_{co} \\ AP_{j,W}^{\max} \times \frac{ws_{j,t,s,d,m} - ws_{ci}}{ws_{no} - ws_{ci}} & \text{if } ws_{ci} \leq ws_{j,t,s,d,m} \leq ws_{no} \\ AP_{j,W}^{\max} & \text{if } ws_{no} < ws_{j,t,s,d,m} \leq ws_{co} \end{cases} \quad (36)$$

$$AP_{j,t,s,d,m}^{PV} = \begin{cases} AP_{j,PV}^{\max} \times \left(\frac{sr_{j,t,s,d,m}^2}{sr_C \times sr_{std}} \right), & 0 \leq sr_{j,t,s,d,m} \leq sr_C \\ AP_{j,PV}^{\max} \times \left(\frac{sr_{j,t,s,d,m}}{sr_{std}} \right), & sr_C < sr_{j,t,s,d,m} \leq sr_{std} \\ AP_{j,PV}^{\max}, & sr_{std} < sr_{j,t,s,d,m} \end{cases} \quad (37)$$

$$A_{j,t,s,d,m}^W = AP_{j,t,s,d,m}^W / \alpha_j^W \quad (38)$$

$$A_{j,t,s,d,m}^{PV} = AP_{j,t,s,d,m}^{PV} / \alpha_j^{PV} \quad (39)$$

$$RP_{j,t,s,d,m}^W = A_{j,t,s,d,m}^W \times \sin\left(\cos^{-1}\left(\alpha_j^W\right)\right) \quad (40)$$

$$RP_{j,t,s,d,m}^{PV} = A_{j,t,s,d,m}^{PV} \times \sin\left(\cos^{-1}\left(\alpha_j^{PV}\right)\right) \quad (41)$$

3.3. PEV constraints

This subsection mathematically formulates the constraints applied to PEVs. Each PEV is characterized by four uncertain variables: arrival time, departure time, arrival SOC, and departure SOC. These variables are derived using the MCS and BRM based on historical records, where a truncated PDF is employed to represent them within their specified limits, as depicted in equations (42) to (45).

$$\left(\forall j \in \Phi_{(PEV-PL)} = [1, \delta_{(PEV-PL)}^{number}], \forall i \in \Phi_{PEV}^j = [1, \delta_{j,(PEV-PL)}^{size}], \forall s \in \Phi_S, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M \right)$$

$$t_{i,s,d,m}^{arr,PEV} = f_{TG} \left(x; \mu_{i,time}^{arr,PEV}, \left(\sigma_{i,time}^{arr,PEV} \right)^2, \left(t_{i,min}^{arr,PEV}, t_{i,max}^{arr,PEV} \right) \right) \quad (42)$$

$$t_{i,s,d,m}^{dep,PEV} = f_{TG} \left(x; \mu_{i,time}^{dep,PEV}, \left(\sigma_{i,time}^{dep,PEV} \right)^2, \left(t_{i,min}^{dep,PEV}, t_{i,max}^{dep,PEV} \right) \right) \quad (43)$$

$$SOC_{i,s,d,m}^{arr,PEV} = f_{TG} \left(x; \mu_{i,SOC}^{arr,PEV}, \left(\sigma_{i,SOC}^{arr,PEV} \right)^2, \left(SOC_{i,min}^{arr,PEV}, SOC_{i,max}^{arr,PEV} \right) \right) \quad (44)$$

$$SOC_{i,s,d,m}^{dep,PEV} = f_{TG} \left(x; \mu_{i,SOC}^{dep,PEV}, \left(\sigma_{i,SOC}^{dep,PEV} \right)^2, \left(SOC_{i,min}^{dep,PEV}, SOC_{i,max}^{dep,PEV} \right) \right) \quad (45)$$

The arriving and departing energy for each PEV can be determined in (46) and (47) based on the predefined uncertain arriving and departing SOC and the known maximum energy capacity of each PEV. Additionally, the minimum energy of each PEV can be determined in (48) based on the specified minimum SOC and the known maximum energy capacity. Conversely, the rated energy of each PEV can be defined in (49) based on the specified maximum SOC and the known maximum energy capacity. Moreover, the maximum active power for each PEV is calculated in (50), depending on the known maximum apparent power of each PEV and the assigned power factor. Furthermore, the rated active power for each PEV must be constrained to the computed maximum

PEV to enter either charging or discharging mode is its presence within the PEV-PL, i.e., when the current time t falls within the specified arrival and departure times of the PEV. The second condition for a PEV to enter charging mode is that its departure or current energy level surpasses the arrival energy level. Conversely, for a PEV to enter discharging mode, the assigned arrival or current energy level must exceed the assigned departure energy.

Furthermore, determining the active power value for a PEV during charging or discharging mode is contingent upon the constrained charging or discharging rate decision variable, as illustrated in (54), in conjunction with the arrival and departure energies. This meticulously crafted control strategy ensures that each PEV attains its desired energy level at departure. Moreover, it guarantees that the PEV's battery is utilized for only half of its cycle. This strategic approach prolongs the lifespan of the PEV's battery and mitigates the need for additional payments to the PEV owner arising from battery degradation resulting from frequent usage.

$$\left(\forall j \in \Phi_{(PEV-PL)} = [1, \delta_{(PEV-PL)}^{number}], \forall i \in \Phi_{PEV}^j = [1, \delta_{j,(PEV-PL)}^{size}], \forall s \in \Phi_S, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M \right)$$

$$AP_{i,t,s,d,m}^{ch,PEV} = \begin{cases} \delta_{i,t,s,d,m}^{ch,PEV} \times (EG_{i,s,d,m}^{dep,PEV} - EG_{i,t,s,d,m}^{arr,PEV}), & \left\{ \begin{array}{l} \text{if } t \in (t_{i,s,d,m}^{arr,PEV}, t_{i,s,d,m}^{dep,PEV}) \text{ \& } \\ \text{if } EG_{i,s,d,m}^{dep,PEV} > EG_{i,t,s,d,m}^{PEV} > EG_{i,s,d,m}^{arr,PEV} \end{array} \right\} \\ 0 & \text{Otherwise} \end{cases} \quad (52)$$

$$AP_{i,t,s,d,m}^{disch,PEV} = \begin{cases} \delta_{i,t,s,d,m}^{disch,PEV} \times (EG_{i,s,d,m}^{arr,PEV} - EG_{i,t,s,d,m}^{dep,PEV}), & \left\{ \begin{array}{l} \text{if } t \in (t_{i,s,d,m}^{arr,PEV}, t_{i,s,d,m}^{dep,PEV}) \text{ \& } \\ \text{if } EG_{i,s,d,m}^{arr,PEV} > EG_{i,t,s,d,m}^{PEV} > EG_{i,s,d,m}^{dep,PEV} \end{array} \right\} \\ 0 & \text{Otherwise} \end{cases} \quad (53)$$

active power, as depicted in (51).

$$EG_{i,s,d,m}^{arr,PEV} = SOC_{i,s,d,m}^{arr,PEV} \times EG_{i,PEV}^{\max} \quad (46)$$

$$EG_{i,s,d,m}^{dep,PEV} = SOC_{i,s,d,m}^{dep,PEV} \times EG_{i,PEV}^{\max} \quad (47)$$

$$EG_{i,PEV}^{\min} = SOC_{i,PEV}^{\min} \times EG_{i,PEV}^{\max} \quad (48)$$

$$EG_{i,PEV}^{\text{rated}} = SOC_{i,PEV}^{\max} \times EG_{i,PEV}^{\max} \quad (49)$$

$$AP_{i,PEV}^{\max} = A_{i,PEV}^{\max} \times \alpha_i^{PEV} \quad (50)$$

$$AP_{i,PEV}^{\text{rated}} \leq AP_{i,PEV}^{\max} \quad (51)$$

3.4. PEV charging/discharging control strategy

This subsection outlines the rigorous mathematical formulation of the charging and discharging control strategy for the parked PEVs. The prescribed charging and discharging active power for each PEV is delineated in (52) and (53). It's evident that the first prerequisite for a

$$0 \leq \left(\delta_{i,t,s,d,m}^{ch,PEV}, \delta_{i,t,s,d,m}^{disch,PEV} \right) \leq \frac{AP_{i,PEV}^{\text{rated}}}{EG_{i,PEV}^{\text{rated}} - EG_{i,PEV}^{\min}} \quad (54)$$

The computed charging and discharging active power must be constrained by the rated active power value, as demonstrated in (55) and (56). Additionally, the formulations presented in (57)-(59) are introduced to prevent simultaneous charging and discharging for the PEV. Moreover, the reactive power provided by each PEV can be computed in (60) based on the calculated value of the active power during charging or discharging modes and the assigned power factor.

$$AP_{i,t,s,d,m}^{ch,PEV} \leq AP_{i,PEV}^{\text{rated}} \quad (55)$$

$$AP_{i,t,s,d,m}^{disch,PEV} \leq AP_{i,PEV}^{\text{rated}} \quad (56)$$

$$AP_{i,t,s,d,m}^{ch,PEV} \leq \delta_{i,t,s,d,m}^{ch,PEV} \times AP_{PEV} \quad (57)$$

$$AP_{i,t,s,d,m}^{disch,PEV} \leq \lambda_{i,t,s,d,m}^{disch,PEV} \times AP_{PEV} \quad (58)$$

$$\lambda_{i,t,s,d,m}^{ch,PEV} + \lambda_{i,t,s,d,m}^{disch,PEV} \leq 1 \quad (59)$$

$$RP_{i,t,s,d,m}^{disch,PEV} = AP_{i,t,s,d,m}^{(ch,disch),PEV} \times \tan(\cos^{-1}(\alpha_i^{PEV})) \quad (60)$$

The energy level of the PEV for the next hour can be updated according to (61), considering the charging, discharging, and floating modes. Updating the energy level during charging or discharging mode necessitates the current energy value, the imposed charged or discharged active power value, the charging or discharging efficiency, and the time horizon set to one hour. The charging, discharging, and floating modes conditions are elaborated in (61).

$$\left(\forall j \in \Phi_{(PEV-PL)} = [1, \delta_{(PEV-PL)}^{number}], \forall i \in \Phi_{PEV}^j = [1, \delta_{j,(PEV-PL)}^{size}], \forall s \in \Phi_S, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M \right)$$

$$EG_{j,(PEV-PL)}^{\max} = \delta_{j,(PEV-PL)}^{size} \times EG_{i,PEV}^{\max} \quad (66)$$

$$A_{j,(PEV-PL)}^{\max} = \delta_{j,(PEV-PL)}^{size} \times A_{i,PEV}^{\max} \quad (67)$$

In (68), a binary variable, denoted as EV , is introduced. It records a value of '1' if a specific PEV within the fleet of PEV-PLs is currently parked in the PEV-PL; otherwise, it is set to '0'. This binary variable is instrumental in computing the current number of parked PEVs in the PEV-PL, as depicted in (69). It's essential to emphasize that the total number of parked PEVs in the PEV-PL must not exceed the obtained size of the corresponding PEV-PL, as presented in (70).

$$EG_{i,t+1,s,d,m}^{PEV} = \begin{cases} EG_{i,t,s,d,m}^{PEV} + \frac{P_{ch}^{PEV}}{\eta_{ch}^{PEV}} \times \Delta t \times AP_{i,t,s,d,m}^{ch,PEV}, & \left\{ \begin{array}{l} \text{if } t \in (t_{i,s,d,m}^{arr,PEV}, t_{i,s,d,m}^{dep,PEV}) \& \\ (EG_{i,t,s,d,m}^{dep,PEV} > EG_{i,t,s,d,m}^{PEV} > EG_{i,t,s,d,m}^{arr,PEV}) \end{array} \right\}, \text{charging} \\ EG_{i,t,s,d,m}^{PEV} - \frac{\Delta t \times AP_{i,t,s,d,m}^{disch,PEV}}{\eta_{disch}^{PEV}}, & \left\{ \begin{array}{l} \text{if } t \in (t_{i,s,d,m}^{arr,PEV}, t_{i,s,d,m}^{dep,PEV}) \& \\ (EG_{i,t,s,d,m}^{arr,PEV} > EG_{i,t,s,d,m}^{PEV} > EG_{i,t,s,d,m}^{dep,PEV}) \end{array} \right\}, \text{discharging} \\ EG_{i,t,s,d,m}^{PEV}, & \left\{ \begin{array}{l} \text{if } t \in (t_{i,s,d,m}^{arr,PEV}, t_{i,s,d,m}^{dep,PEV}) \& \\ (EG_{i,t,s,d,m}^{arr,PEV} = EG_{i,t,s,d,m}^{dep,PEV} \parallel \\ EG_{i,t,s,d,m}^{PEV} = EG_{i,t,s,d,m}^{dep,PEV}) \end{array} \right\}, \text{Floating} \end{cases} \quad (61)$$

The constraints on the energy level at any time t during charging and discharging modes, the energy must be confined within the specified arrival and departure energy levels, as demonstrated in (62) and (63). Furthermore, it's imperative to note that this energy level must always be bounded by the minimum and rated energy levels, as outlined in (64). Moreover, the hourly apparent power value for each PEV can be computed in (65), depending on the hourly charging or discharging active power and the hourly discharged reactive power.

$$EG_{i,t,s,d,m}^{arr,PEV} \leq EG_{i,t,s,d,m}^{PEV} \leq EG_{i,t,s,d,m}^{dep,PEV} \text{ if } EG_{i,t,s,d,m}^{arr,PEV} < EG_{i,t,s,d,m}^{dep,PEV} \quad (62)$$

$$EG_{i,t,s,d,m}^{dep,PEV} \leq EG_{i,t,s,d,m}^{PEV} \leq EG_{i,t,s,d,m}^{arr,PEV} \text{ if } EG_{i,t,s,d,m}^{arr,PEV} > EG_{i,t,s,d,m}^{dep,PEV} \quad (63)$$

$$EG_{i,PEV}^{\min} \leq EG_{i,t,s,d,m}^{PEV} \leq EG_{i,PEV}^{\text{rated}} \quad (64)$$

$$A_{i,t,s,d,m}^{PEV} = \sqrt{(AP_{i,t,s,d,m}^{(ch,disch),PEV})^2 + (RP_{i,t,s,d,m}^{disch,PEV})^2} \quad (65)$$

3.5. PEV-PL constraints

This subsection presents the decisive mathematical outcomes derived from sections 3.3 and 3.4. The maximum energy capacity and apparent power for the deployed PEV-PLs can be computed in (66) and (67). These values play a pivotal role in determining the associated investment cost. These maximum capacities are contingent upon the obtained optimal size of the PEV-PL, which signifies the maximum number of PEVs that can be parked simultaneously within the PEV-PL, along with the maximum energy and apparent power capacities of each PEV within this fleet.

$$EV_{i,t,s,d,m}^j = \begin{cases} 1 & \text{if } t_{i,s,d,m}^{arr,PEV} \leq t \leq t_{i,s,d,m}^{dep,PEV} \\ 0 & \text{Otherwise} \end{cases} \quad (68)$$

$$N_{j,t,s,d,m}^{PEV} = \sum_i EV_{i,t,s,d,m}^j \quad (69)$$

$$N_{j,t,s,d,m}^{PEV} \leq \delta_{j,PEV-PL}^{size} \quad (70)$$

The energy at the time t being sold from the PEV-PL to the currently parked PEVs undergoing charging mode is computed in (71). Conversely, (72) calculates the energy at time t being bought by the PEV-PL from the currently parked PEVs undergoing discharging mode. These values are paramount in determining the total energy expenses and profits corresponding to the deployed PEV-PL. Moreover, (73) computes the reactive power delivered from the PEVs to the PEV-PLs, assuming no tariff is imposed on the exchange of reactive power between PEVs and PEV-PLs, but only on the exchange of active power. Hence, (74) calculates the magnitude of the hourly apparent power for each PEV-PL, which is instrumental in calculating the PEV-PL's operational cost. Finally, (75) determines the currently available energy in the corresponding PEV-PL at the time t , based on the currently available PEVs and their respective current energy levels.

$$AP_{j,t,s,m,y}^{sell,(PEV-PL)} = \sum_i EV_{i,t,s,d,m}^j \times AP_{i,t,s,d,m}^{ch,PEV} \quad (71)$$

$$AP_{j,t,s,m,y}^{buy,(PEV-PL)} = \sum_i EV_{i,t,s,d,m}^j \times AP_{i,t,s,d,m}^{disch,PEV} \quad (72)$$

$$RP_{j,t,s,d,m}^{(PEV-PL)} = \sum_i EV_{i,t,s,d,m}^j \times RP_{i,t,s,d,m}^{disc,PEV} \quad (73)$$

$$A_{j,t,s,m,y}^{(PEV-PL)} = \sqrt{\left(AP_{j,t,s,m,y}^{sell/buy,(PEV-PL)}\right)^2 + \left(RP_{j,t,s,m,y}^{(PEV-PL)}\right)^2} \quad (74)$$

$$EG_{j,t,s,m,y}^{(PEV-PL)} = \sum_i EV_{i,t,s,d,m}^j \times EG_{i,t,s,d,m}^{PEV} \quad (75)$$

3.6. SBESS/MBESS constraints

This subsection rigorously outlines the constraints governing the deployed SBESS/MBESS through precise mathematical formulations. The determination of the maximum energy and apparent power capacity of each installed SBESS/MBESS is elucidated in (76) and (77), predicated on the optimized size and the inherent maximum energy and apparent power capacity of individual battery units within the SBESS/MBESS. These derived metrics are pivotal in assessing the associated investment and maintenance costs for the respective SBESS/MBESS. Additionally, (78) elucidates the computation of the maximum active power for each SBESS/MBESS, contingent upon the computed maximum apparent power and the designated power factor. Furthermore, determining the maximum permissible values of reactive power for SBESS/MBESS is illustrated in (79). Moreover, determining the minimum and rated energy capacities of the SBESS/MBESS is achievable through calculations hinged upon the attained maximum capacity and the designated minimum and maximum SOC, delineated in (80) and (81), respectively. The inclusion of formulations (82) and (83) guarantees parity in the energy level of the SBESS/MBESS at the onset and culmination of each day, thereby ensuring consistency and aligning with the initial stored energy at the point of installation.

$$\left(\forall j \in \Phi_{(SBESS,MBESS)} = \left[1, \delta_{(SBESS,MBESS)}^{number}\right], \forall i \in \Phi_{(SBatt,MBatt)}^j \left[1, \delta_{j,(SBESS,MBESS)}^{size}\right], \forall s \in \Phi_s, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M\right)$$

$$EG_{j,BESS}^{\max} = \delta_{j,BESS}^{size} \times EG_{i,Batt}^{\max} \quad (76)$$

$$A_{j,BESS}^{\max} = \delta_{j,BESS}^{size} \times A_{i,Batt}^{\max} \quad (77)$$

$$AP_{j,BESS}^{\max} = A_{j,BESS}^{\max} \times \alpha_j^{BESS} \quad (78)$$

$$RP_{j,BESS}^{\max} = \sqrt{\left(A_{j,BESS}^{\max}\right)^2 - \left(AP_{j,BESS}^{\max}\right)^2} \quad (79)$$

$$EG_{j,BESS}^{\min} = SOC_{j,BESS}^{\min} \times EG_{j,BESS}^{\max} \quad (80)$$

$$E_{j,BESS}^{\text{rated}} = SOC_{j,BESS}^{\max} \times E_{j,BESS}^{\max} \quad (81)$$

$$EG_{j,t,s,d,m}^{BESS}(t=1) = EG_{j,BESS}^{\text{initial}} \quad (82)$$

$$EG_{j,t,s,d,m}^{BESS}(t=T) = EG_{j,BESS}^{\text{initial}} \quad (83)$$

3.7. SBESS/MBESS charging/discharging control strategy

This subsection systematically formulates an optimal and adaptable charging and discharging control strategy tailored for the deployed SBESSs/MBESSs. Leveraging both present and past states $(-1, 0, +1)$ of the designated decision variable, symbolic of the SBESS/MBESS status,

introduces a novel binary variable λ expounded upon in (84)-(86). This binary counterpart plays a pivotal role in orchestrating the operational modalities – encompassing charging, discharging, and floating – of the SBESSs/MBESSs, enhancing their operational efficacy and adaptability.

$$\left(\forall j \in \Phi_{(SBESS,MBESS)} = \left[1, \delta_{(SBESS,MBESS)}^{number}\right], \forall s \in \Phi_s, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M\right)$$

$$\lambda_{j,t,s,d,m}^{ch,BESS} = \begin{cases} 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = -1 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = 1\right) \\ 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = 0 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = 1\right) \\ 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = 1 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = 1\right) \\ 0 & \text{otherwise} \end{cases} \quad (84)$$

$$\lambda_{j,t,s,d,m}^{disch,BESS} = \begin{cases} 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = 1 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = -1\right) \\ 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = 0 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = -1\right) \\ 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = -1 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = -1\right) \\ 0 & \text{otherwise} \end{cases} \quad (85)$$

$$\lambda_{j,t,s,d,m}^{floating,BESS} = \begin{cases} 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = 1 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = 0\right) \\ 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = -1 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = 0\right) \\ 1 & \text{if } \left(\delta_{j,t-1,s,d,m}^{status,BESS} = 0 \rightarrow \delta_{j,t,s,d,m}^{status,BESS} = 0\right) \\ 0 & \text{otherwise} \end{cases} \quad (86)$$

The cumulative variation between preceding and current binary values derived from (84)-(86) for charging and discharging modes must not surpass twice the designated number of cycles allotted for each SBESS/MBESS, typically set to one, detailed in (87) and (88). These formulaic constraints enforce a singular cycle per day for each SBESS/MBESS, a strategic measure aimed at prolonging the lifespan of the energy storage system and averting the necessity for costly replacements in the long run. Furthermore, this approach can mitigate short- and long-term operational expenses, ultimately reducing overall expenditure.

$$\sum_{t=1}^T \left| \lambda_{j,t,s,d,m}^{ch,BESS} - \lambda_{j,t-1,s,d,m}^{ch,BESS} \right| \leq 2 \times N_{cyc}^{BESS} \quad (87)$$

$$\sum_{t=1}^T \left| \lambda_{j,t,s,d,m}^{disch,BESS} - \lambda_{j,t-1,s,d,m}^{disch,BESS} \right| \leq 2 \times N_{cyc}^{BESS} \quad (88)$$

Furthermore, it is imperative that the accumulated disparity between preceding and current binary values, as determined by (84)-(86) representing charging, discharging, and floating modes, surpasses zero for each SBESS/MBESS, as depicted in (89)-(91). This prerequisite guarantees the occurrence of charging, discharging, and floating modes without fail daily. Additionally, the initial values of the binary variables governing charging and discharging, derived from (84)-(86), are initially configured to '0', as illustrated in (92). Moreover, formulations delineated in (93)-(97) are strategically introduced to prevent the occurrence of concurrent charging, discharging, and floating operations for the SBESS/MBESS.

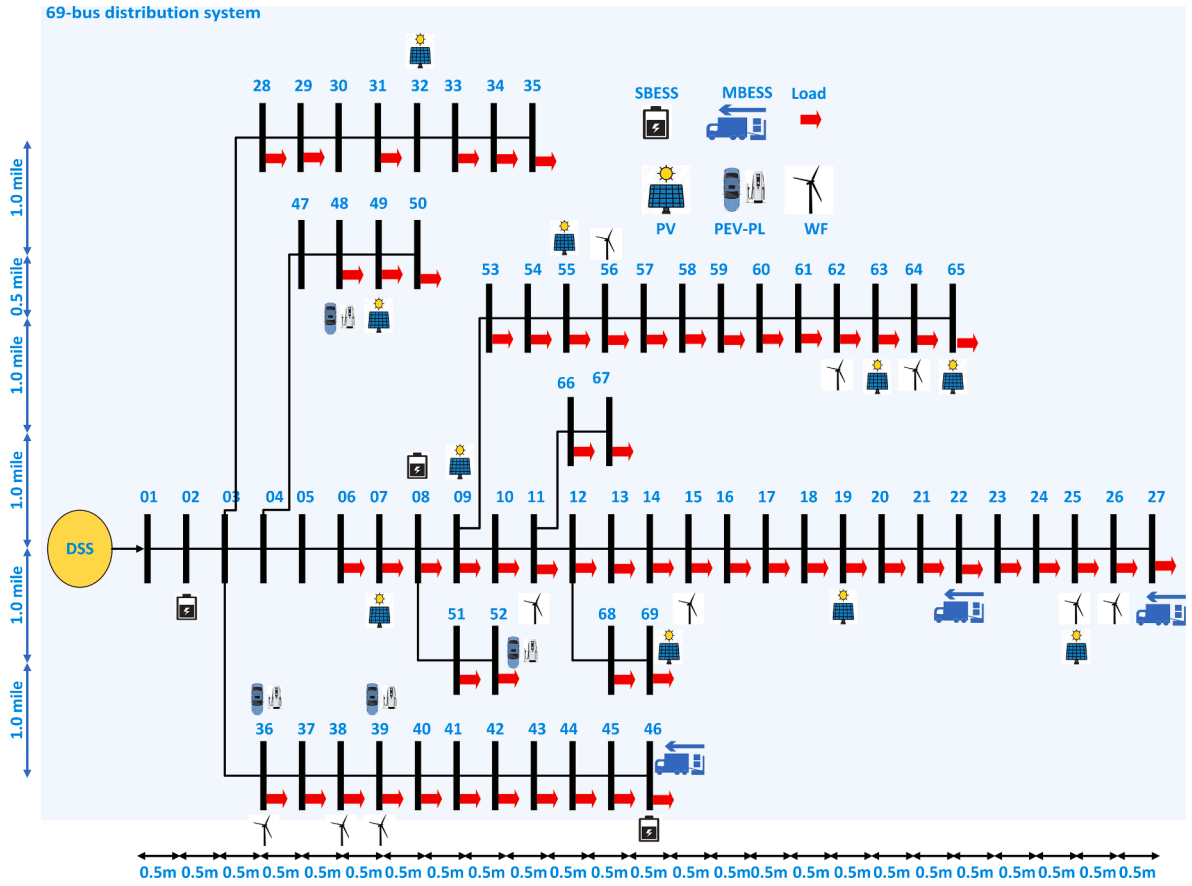


Fig. 2. 69 bus distribution system with the optimally allocated components (wind/PV/PEV-PL/SBESS-MBESS) in July considering case 5.

Table 1

Input technical and economic parameters of 69 bus-DS.

Parameters	values
BMVA, BKV	10MVA, 12.66KV
N_{DSS}	1
N_{brch}	68
N_{bus}	69
$AP_{j,DSS}^{max}$ (MW)	
$AP_{j,DSS}^{max}$ (MVAR)	
V_j^{min}, V_j^{max} (pu)	0.95–1.05
f_{DSS}^{Em} (\$/MWh)	22.5 [62]

Table 2

Input technical and economic parameters of wind DGs and PV DGs.

Parameters	values
N_W^{max}, N_{PV}^{max}	10
$N_{Wu}^{max}, N_{PVu}^{max}$	5
$A_{i,Wu}^{max}, A_{i,PVu}^{max}$ (KVA)	50
$\alpha_j^W, \alpha_j^{PV}$	0.95
$f_W^{invest}, f_{PV}^{invest}$ (\$/KVA)	1250, 1100
$f_W^{oper}, f_{PV}^{oper}$ (\$/MWh)	2.3, 1.9
Lf_W, Lf_{PV} (years)	20, 30
$ws_{cl}, ws_{no}, ws_{co}$ (m/s)	4, 14, 25
IR	5/100
IF	1/100

Table 3

Input economic and technical parameters of PEV-PLs and PEVs.

Parameters	values
$N_{(PEV-PL)}^{max}$	5
N_{PEVs}^{max}	100
$\xi_{ch}^{PEV}, \xi_{disch}^{PEV}$	95 %
Δt (hour)	1
α_i^{PEV}	0.95
$SOC_{i,PEV}^{min}, SOC_{i,PEV}^{max}$	0.3, 0.9
$EG_{i,PEV}^{max}$ (kWh)	42 [44]
$A_{i,PEV}^{max}$ (KVA)	13.2 [44]
$f_{(PEV-PL)}^{EG,max}$ (\$/kWh)	225
$f_{(PEV-PL)}^{A,max}$ (\$/KVA)	175
$f_{(PEV-PL)}^{oper}$ (\$/MWh)	1.9
$f_{(PEV-PL)}^{Tarrif-buy}$	$J_{t,s,d,m}^{EP}$
$f_{(PEV-PL)}^{Tarrif-sell}$	$1.1 \times J_{t,s,d,m}^{EP}$
$Lf_{(PEV-PL)}$ (years)	10

$$\sum_{t=1}^T \left| \lambda_{j,t,s,d,m}^{ch,BESS} - \lambda_{j,t-1,s,d,m}^{ch,BESS} \right| > 0 \quad (89)$$

$$\sum_{t=1}^T \left| \lambda_{j,t,s,d,m}^{disch,BESS} - \lambda_{j,t-1,s,d,m}^{disch,BESS} \right| > 0 \quad (90)$$

$$\sum_{t=1}^T \left| \lambda_{j,t,s,d,m}^{floating,BESS} - \lambda_{j,t-1,s,d,m}^{floating,BESS} \right| > 0 \quad (91)$$

$$\lambda_{j,t,s,d,m}^{ch,BESS}(t=0) = \lambda_{j,t,s,d,m}^{disch,BESS}(t=0) = 0 \quad (92)$$

Table 4
Input technical and economic parameters of SBESS and MBESS.

Parameters	values
N_{BESS}^{max} (SBESS/MBESS)	5
N_{Batu}^{max}	5
$\epsilon_{ch}^{BESS}, \epsilon_{disch}^{BESS}$	95 %
Δt (hour)	1
α_j^{BESS}	0.95
$SOC_{LBESS}^{initial}, SOC_{LBESS}^{min}, SOC_{LBESS}^{max}$	0.7, 0.3, 0.9
$EG_{i,Batu}^{max}$ (kWh)	90
$A_{i,Batu}^{max}$ (KVA)	18
f_{BESS}^{EGmax} (\$/kWh)	225
f_{BESS}^{max} (\$/KW)	175
f_{BESS}^{oper} (\$/MWh)	1.5
f_{fix}^{trip} (\$/trip)	100
f_{Var}^{trip} (\$/mile/trip)	0.5
Lf_{BESS} (years)	10

Table 5
PEV's probability distribution.

Parameter	μ_i	σ_i	Min	Max
$t_i^{arr,PEV}$	3:00	8:00	5:00	17:00
$t_i^{dep,PEV}$	18:00	3:00	16:00	24:00
$SOC_i^{arr,PEV}$	0.5	0.25	0.3	0.9
$SOC_i^{dep,PEV}$	0.7	0.25	0.6	0.9

$$AP_{j,t,s,d,m}^{ch,BESS} \leq \lambda_{j,t,s,d,m}^{ch,BESS} \times AP_{BESS} \quad (93)$$

$$AP_{j,t,s,d,m}^{disch,BESS} \leq \lambda_{j,t,s,d,m}^{disch,BESS} \times AP_{BESS} \quad (94)$$

$$\lambda_{j,t,s,d,m}^{ch,BESS} + \lambda_{j,t,s,d,m}^{disch,BESS} \leq 1 \quad (95)$$

$$\lambda_{j,t,s,d,m}^{ch,BESS} + \lambda_{j,t,s,d,m}^{floating,BESS} \leq 1 \quad (96)$$

$$\lambda_{j,t,s,d,m}^{disch,BESS} + \lambda_{j,t,s,d,m}^{floating,BESS} \leq 1 \quad (97)$$

Determining charging and discharging active power for each SBESS/MBESS is facilitated through (98) and (99). This computation intricately hinges upon several factors, including the rate of charging and discharging decision variables constrained by (100), the binary values derived from (84) and (85), the rated energy capacity outlined in (81),

and the current energy value of the SBESS/MBESS. It's worth emphasizing that (100) is included to uphold the fulfilment of constraint (101), which mandates that charging and discharging active power remain within prescribed rated and maximum thresholds.

$$AP_{j,t,s,d,m}^{ch,BESS} = \lambda_{j,t,s,d,m}^{ch,BESS} \times \delta_{j,t,s,d,m}^{ch,BESS} \times (EG_{j,BESS}^{rated} - EG_{j,t,s,d,m}^{BESS}) \quad (98)$$

$$AP_{j,t,s,d,m}^{disch,BESS} = \lambda_{j,t,s,d,m}^{disch,BESS} \times \delta_{j,t,s,d,m}^{disch,BESS} \times (EG_{j,t,s,d,m}^{BESS} - EG_{j,BESS}^{min}) \quad (99)$$

$$0 \leq (\delta_{j,t,s,d,m}^{ch,BESS}, \delta_{j,t,s,d,m}^{disch,BESS}) \leq \frac{AP_{j,BESS}^{rated}}{EG_{j,BESS}^{rated} - EG_{j,BESS}^{min}} \quad (100)$$

$$AP_{j,t,s,d,m}^{(ch/dis)ch,BESS} \leq AP_{j,BESS}^{rated} < AP_{j,BESS}^{max} \quad (101)$$

The energy state of the SBESS/MBESS for the subsequent hour can be efficiently revised by (102) meticulously considering the dynamic interplay of charging, discharging, and floating modes. When updating the energy level coincides with charging or discharging operations, a meticulous consideration of the current energy status, the applied active power for charging or discharging, the efficiency of charging or discharging, and a predefined time frame of one hour becomes imperative. The criteria delineating the occurrence of charging, discharging, and floating modes are elaborately articulated within (102), intricately contingent upon the binary variables derived from (84)-(86).

$$EG_{j,t+1,s,d,m}^{BESS} = \begin{cases} EG_{j,t,s,d,m}^{BESS} + \epsilon_{ch}^{BESS} \times \Delta t \times AP_{j,t,s,d,m}^{ch,BESS} & \text{if } \lambda_{j,t,s,d,m}^{ch,BESS} = 1 \\ EG_{j,t,s,d,m}^{BESS} - \frac{\Delta t \times AP_{j,t,s,d,m}^{disch,BESS}}{\epsilon_{disch}^{BESS}} & \text{if } \lambda_{j,t,s,d,m}^{disch,BESS} = 1 \\ EG_{j,t,s,d,m}^{BESS} & \text{if } \lambda_{j,t,s,d,m}^{floating,BESS} = 1 \end{cases} \quad (102)$$

Moreover, it is imperative that the updated energy level for the subsequent hour remains within the bounds of the minimum and rated energy values meticulously stipulated in (103). Additionally, the inclusion of (104) is paramount to guaranteeing the daily energy equilibrium within the SBESS/MBESS, thereby satisfying the conditions outlined in (82) and (83).

$$EG_{j,BESS}^{min} \leq EG_{j,t+1,s,d,m}^{BESS} \leq EG_{j,BESS}^{rated} \quad (103)$$

$$\sum_{t=1}^T \epsilon_{ch}^{BESS} AP_{j,t,s,d,m}^{ch,BESS} = \sum_{t=1}^T \frac{AP_{j,t,s,d,m}^{disch,BESS}}{\epsilon_{disch}^{BESS}} \quad (104)$$

Moreover, determining the apparent power for each SBESS/MBESS,

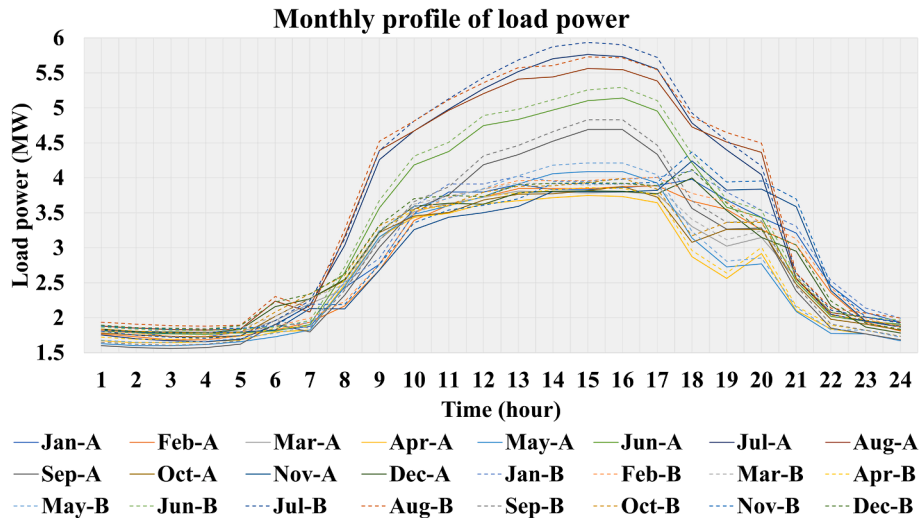


Fig. 3. The monthly profiles of load power considering scenarios A and B.

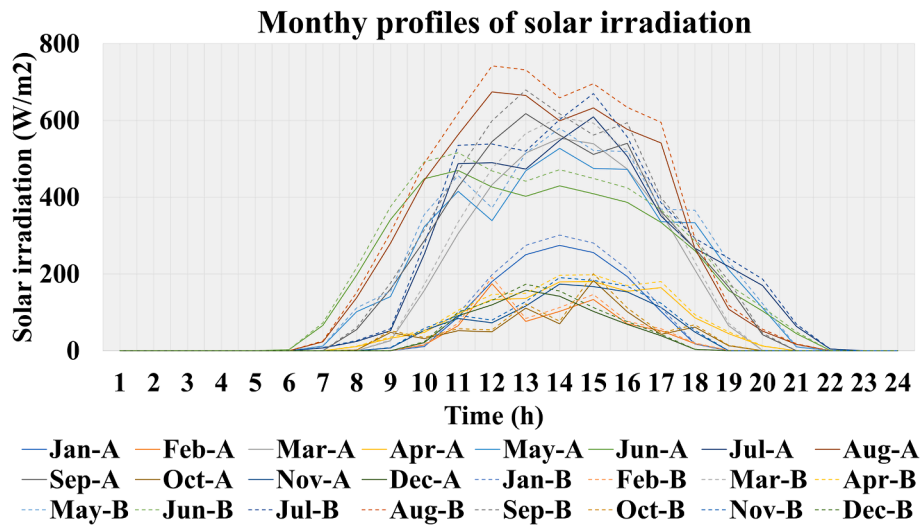


Fig. 4. The monthly profiles of solar irradiation considering scenarios A and B.

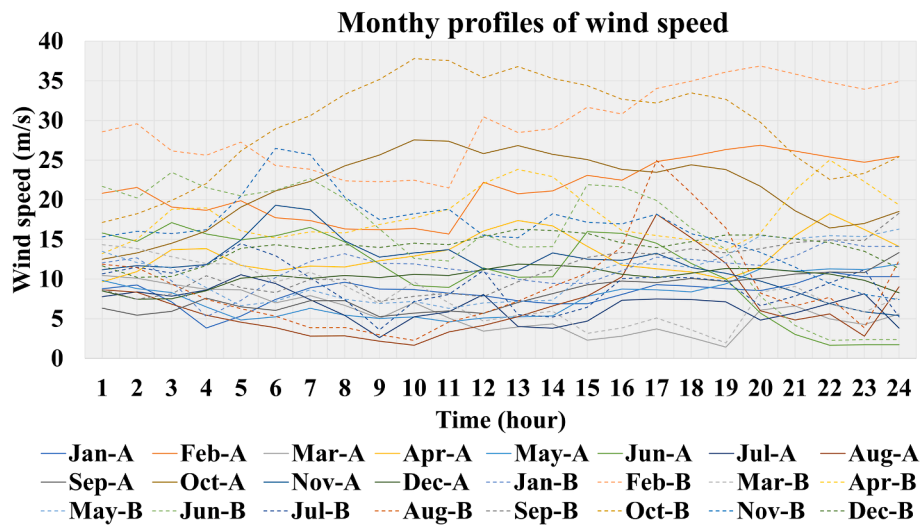


Fig. 5. The monthly profiles of wind speed considering scenarios A and B.

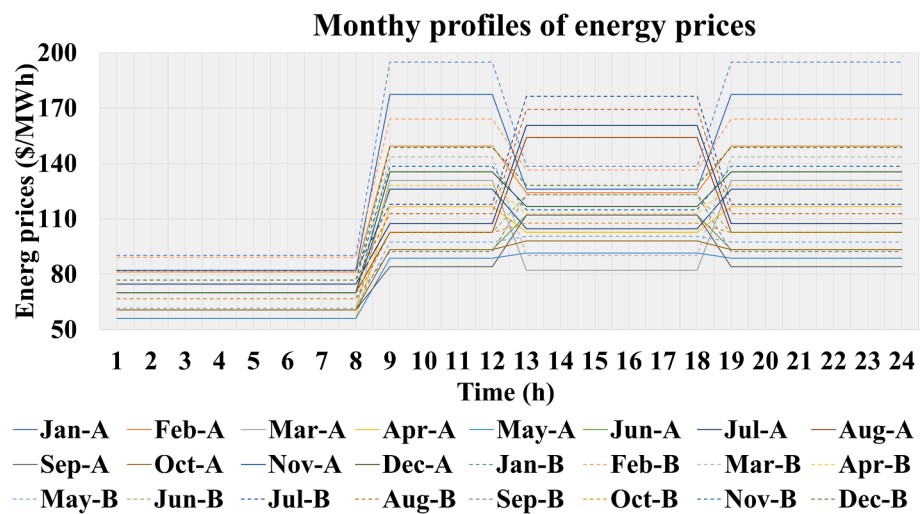


Fig. 6. The monthly profiles of energy prices considering scenarios A and B.

Table 6

The monthly availability range of uncertain variables.

Source	Peak	Valley
Load	Jul A, Jul B, Aug A and Aug B	Apr B, Apr A, May B, May A
Wind speed	Oct B, Feb B, and Apr B	Aug A, Aug B, Mar A, Mar B
Solar irradiation	Aug B, Aug A, Jul B and Sep A	Oct B, Dec B
Energy prices	Jan B, Jan A, Jul B, Aug B	May A, May B, Mar B

crucial for operational cost assessment, is achieved through calculations integrating the charging and discharging active power alongside the designated power factor, as elucidated in (105). Furthermore, the computation of the reactive power contribution attributable to the SBESS/MBESS is outlined in (106).

$$A_{j,t,s,d,m}^{BESS} = AP_{j,t,s,d,m}^{(ch,disch),BESS} / \alpha_j^{BESS} \quad (105)$$

$$RP_{j,t,s,d,m}^{disch,BESS} = \sqrt{\left(A_{j,t,s,d,m}^{BESS}\right)^2 - \left(AP_{j,t,s,d,m}^{(ch,disch),BESS}\right)^2} \quad (106)$$

3.8. Network constraints

This subsection meticulously delineates the mathematical constraints governing the robust and seamless operation of the system's network. To ensure operational integrity, the voltage levels at each bus must strictly adhere to predetermined minimum and maximum thresholds, outlined in (107). Additionally, stringent limits are imposed on active and reactive power for the DSS, meticulously detailed in (108)

Table 7

Optimal locations and sizes considering case 2.

Wind DGs			PV DGs			PEV-PL			
Bus	Units	Size (MVA)	Bus	Units	Size (MVA)	Bus	PEV units	Size	
								MWh	MVA
21	4	0.2	47	2	0.1	35	31	1.302	0.4092
44	2	0.1	2	2	0.1	2	24	1.008	0.3168
31	3	0.15	42	3	0.15	59	38	1.596	0.5016
56	4	0.2	44	4	0.2	59	53	2.226	0.6996
42	2	0.1	65	3	0.15	—	—	—	—
65	5	0.25	41	5	0.25	—	—	—	—
19	5	0.25	63	2	0.1	—	—	—	—
63	5	0.25	31	1	0.05	—	—	—	—
25	5	0.25	—	—	—	—	—	—	—
62	5	0.25	—	—	—	—	—	—	—

Table 8

Optimal locations and sizes considering case 3.

Optimal locations and sizes considering case 3													
Wind DGs			PV DGs			PEV-PL				SBESS			
Bus	Units	Size (MVA)	Bus	Units	Size (MVA)	Bus	PEV units	Size		Bus	Batt units	Size	
								MWh	MVA			MWh	MVA
42	2	0.1	61	5	0.25	33	1	0.042	0.0132	17	4	0.36	0.072
64	5	0.25	65	3	0.15	5	28	1.176	0.3696	54	3	0.27	0.054
61	5	0.25	54	2	0.1	57	2	0.084	0.0264	37	3	0.27	0.054
58	5	0.25	49	5	0.25	52	37	1.554	0.4884	23	2	0.18	0.036
53	5	0.25	40	4	0.2	46	138	5.796	1.8216	24	1	0.09	0.018
15	5	0.25	45	3	0.15	—	—	—	—	—	—	—	—
66	5	0.25	17	5	0.25	—	—	—	—	—	—	—	—
65	5	0.25	—	—	—	—	—	—	—	—	—	—	—
44	2	0.1	—	—	—	—	—	—	—	—	—	—	—
60	5	0.25	—	—	—	—	—	—	—	—	—	—	—

Table 9

Optimal locations and sizes considering case 4.

Optimal locations and sizes considering case 4									
Wind DGs			PV DGs			PEV-PL			
Bus	Units	Size (MVA)	Bus	Units	Size (MVA)	Bus	PEV units	Size	
								MWh	MVA
17	5	0.25	34	2	0.1	44	3	0.126	0.0396
64	5	0.25	47	2	0.1	57	12	0.504	0.1584
61	5	0.25	13	1	0.05	18	13	0.546	0.1716
27	5	0.25	65	5	0.25	57	39	1.638	0.5148
53	5	0.25	58	4	0.2	47	145	6.09	1.914
32	3	0.15	37	2	0.1	—	—	—	—
36	3	0.15	8	3	0.15	—	—	—	—
42	5	0.25	11	2	0.1	—	—	—	—
25	2	0.1	21	4	0.2	—	—	—	—
14	5	0.25	—	—	—	—	—	—	—

and (109), respectively. Furthermore, to maintain network stability, the power flow along each branch must not surpass its rated capacity, a condition rigorously stipulated in (110).

$$(\forall i \in \Phi_{DSS}, \forall j \in \Phi_{DSS}, \forall s \in \Phi_S, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M)$$

$$V_j^{\min} \leq V_{j,t,s,d,m} \leq V_j^{\max} \quad (107)$$

$$AP_{j,DSS}^{\min} \leq AP_{i,t,s,d,m}^{DSS} \leq AP_{j,DSS}^{\max} \quad (108)$$

$$RP_{j,DSS}^{\min} \leq RP_{i,t,s,d,m}^{DSS} \leq RP_{j,DSS}^{\max} \quad (109)$$

$$|A_{ij,t,s,d,m}^{brch}| \leq A_{ij,brch}^{rated} \quad (110)$$

3.9. Power flow constraints

This subsection delineates the mathematical constraints governing power flow dynamics within the system. Ensuring equilibrium, the injected and computed active and reactive power at each bus are equated, meticulously stipulated in (111). The active and reactive power injections at each bus are determined based on the collective contributions from available DSS, wind DGs, PV DGs, PEV-PLs, SBESSs, MBESSs, commercial loads, and residential loads connected to the respective bus, as outlined in (112) and (113).

$$(\forall i \in \Phi_{bus}, \forall i \in \Phi_{bus}, \forall s \in \Phi_S, t \in \Phi_T, d \in \Phi_D, m \in \Phi_M)$$

$$\begin{cases} AP_{i,t,s,d,m}^{injected} = AP_{i,t,s,d,m}^{calculated} \\ RP_{i,t,s,d,m}^{injected} = RP_{i,t,s,d,m}^{calculated} \end{cases} \quad (111)$$

$$\begin{cases} AP_i^{inj} = AP_{i,t,s,d,m}^{DSS} + \tau_i^W \times AP_{i,t,s,d,m}^W + \tau_i^{PV} \times AP_{i,t,s,d,m}^{PV} + \tau_i^{PEV-PL} \times AP_{i,t,s,d,m}^{disch,(PEV-PL)} + \tau_i^{SBESS} \times AP_{i,t,s,d,m}^{disch,SBESS} \\ + \tau_i^{MBESS} \times AP_{i,t,s,d,m}^{disch,MBESS} - \tau_i^{PEV-PL} \times AP_{i,t,s,d,m}^{ch,(PEV-PL)} - \tau_i^{SBESS} \times AP_{i,t,s,d,m}^{ch,SBESS} - \tau_i^{MBESS} \times AP_{i,t,s,d,m}^{ch,MBESS} - AP_{i,t,s,d,m}^{RL} - AP_{i,t,s,d,m}^{CL} \end{cases} \quad (112)$$

$$\begin{cases} RP_i^{inj} = RP_{i,t,s,d,m}^{DSS} + \tau_i^W \times RP_{i,t,s,d,m}^W + \tau_i^{PV} \times RP_{i,t,s,d,m}^{PV} + \tau_i^{PEV-PL} \times RP_{i,t,s,d,m}^{disch,(PEV-PL)} + \tau_i^{SBESS} \times RP_{i,t,s,d,m}^{disch,SBESS} \\ + \tau_i^{MBESS} \times RP_{i,t,s,d,m}^{disch,MBESS} - RP_{i,t,s,d,m}^{RL} - RP_{i,t,s,d,m}^{CL} \end{cases} \quad (113)$$

Introducing a binary variable in (114) streamlines the process by indicating the availability of each component. Furthermore, the active and reactive power flow traversing branches linked to the corresponding bus can be computed via (115) and (116), further refining the network's power distribution dynamics. Furthermore, (117) and (118) depict the system's active and reactive power balance, showcasing the necessity for equilibrium between the aggregated power provision from DSS, wind DGs, PV DGs, PEV-PLs, SBESSs, and MBESSs – and the collective power demand of PEV-PLs, SBESSs, MBESSs, residential loads, commercial loads, along with total power losses within the branches.

Table 10

Monthly optimal locations and sizes of MBESS considering case 4.

Optimal	Month	Optimal locations and sizes of MBESSs considering case 4			
		MBESS1	MBESS2	MBESS3	MBESS4
Bus location	Jan	55	50	40	42
	Feb	26	35	17	40
	Mar	2	11	49	37
	Apr	42	28	13	11
	May	25	22	28	39
	Jun	18	21	24	22
	Jul	28	28	36	13
	Aug	55	34	51	10
	Sep	36	18	12	69
	Oct	52	11	48	34
	Nov	48	15	29	37
	Dec	41	47	26	23
Size	Number of units	1	3	1	4
	MWh	0.09	0.27	0.09	0.36
	MVA	0.018	0.054	0.018	0.072

$$\tau_i^{(W/PV/(PEV-PL)/SBESS/MBESS)} = \begin{cases} 1 & \text{if } \delta_{i,(W/PV/(PEV-PL)/SBESS/MBESS)}^{size} \neq 0 \\ 0 & \text{otherwise} \end{cases} \quad (114)$$

$$AP_{i,t,s,d,m}^{calculated} = \sum_{j \in \Phi_{bus}, i \neq j} \left[|V_{i,t,s,d,m}|^2 G_{ij} - |V_{i,t,s,d,m}| |V_{j,t,s,d,m}| (G_{ij} \cos \varphi_{ij,t,s,d,m} + B_{ij} \sin \varphi_{ij,t,s,d,m}) \right] \quad (115)$$

$$RP_{i,t,s,d,m}^{calculated} = \sum_{j \in \Phi_{bus}, i \neq j} \left[-|V_{i,t,s,d,m}|^2 B_{ij} - |V_{i,t,s,d,m}| |V_{j,t,s,d,m}| (G_{ij} \cos \varphi_{ij,t,s,d,m} - B_{ij} \sin \varphi_{ij,t,s,d,m}) \right] \quad (116)$$

$$\left\{ \begin{aligned} & \sum_{i \in \Phi_{DSS}} AP_{i,t,s,d,m}^{DSS} + \sum_{i \in \Phi_W} AP_{i,t,s,d,m}^W + \sum_{i \in \Phi_{PV}} AP_{i,t,s,d,m}^{PV} + \sum_{i \in \Phi_{PEV-PL}} AP_{i,t,s,d,m}^{disch,(PEV-PL)} + \sum_{i \in \Phi_{SBESS}} AP_{i,t,s,d,m}^{disch,SBESS} \\ & + \sum_{i \in \Phi_{MBESS}} AP_{i,t,s,d,m}^{disch,MBESS} - \sum_{i \in \Phi_{PEV-PL}} AP_{i,t,s,d,m}^{ch,(PEV-PL)} - \sum_{i \in \Phi_{SBESS}} AP_{i,t,s,d,m}^{ch,SBESS} - \sum_{i \in \Phi_{MBESS}} AP_{i,t,s,d,m}^{ch,MBESS} - \sum_{i \in \Phi_{RL}} AP_{i,t,s,d,m}^{RL} \\ & - \sum_{i \in \Phi_{CL}} AP_{i,t,s,d,m}^{CL} - \sum_{ij \in \Phi_{bus}, i \neq j} AP_{ij,t,s,d,m}^{Loss} = 0 \end{aligned} \right\} \quad (117)$$

$$\left\{ \begin{aligned} & \sum_{i \in \Phi_{DSS}} RP_{i,t,s,d,m}^{DSS} + \sum_{i \in \Phi_W} RP_{i,t,s,d,m}^W + \sum_{i \in \Phi_{PV}} RP_{i,t,s,d,m}^{PV} + \sum_{i \in \Phi_{PEV-PL}} RP_{i,t,s,d,m}^{disch,(PEV-PL)} + \sum_{i \in \Phi_{SBESS}} RP_{i,t,s,d,m}^{disch,SBESS} \\ & + \sum_{i \in \Phi_{MBESS}} RP_{i,t,s,d,m}^{disch,MBESS} - \sum_{i \in \Phi_{RL}} RP_{i,t,s,d,m}^{RL} - \sum_{i \in \Phi_{CL}} RP_{i,t,s,d,m}^{CL} - \sum_{ij \in \Phi_{bus}, i \neq j} RP_{ij,t,s,d,m}^{Loss} = 0 \end{aligned} \right\} \quad (118)$$

Table 11
Optimal locations and sizes considering case 5.

Optimal locations and sizes considering case 5													
Wind DGs			PV DGs			PEV-PL				SBESS			
Bus	Units	Size (MVA)	Bus	Units	Size (MVA)	Bus	Units	Size		Bus	Units	Size	
								MWh	MVA			MWh	MVA
39	2	0.1	32	1	0.05	52	2	0.084	0.0264	8	2	0.18	0.036
56	3	0.15	25	5	0.25	48	21	0.882	0.2772	46	1	0.09	0.018
36	2	0.1	19	5	0.25	36	54	2.268	0.7128	2	4	0.36	0.072
25	4	0.2	49	4	0.2	39	82	3.444	1.0824	—	—	—	—
38	5	0.25	55	4	0.2	—	—	—	—	—	—	—	—
64	3	0.15	9	2	0.1	—	—	—	—	—	—	—	—
11	2	0.1	65	3	0.15	—	—	—	—	—	—	—	—
26	3	0.15	63	5	0.25	—	—	—	—	—	—	—	—
15	5	0.25	7	2	0.1	—	—	—	—	—	—	—	—
62	2	0.1	69	4	0.2	—	—	—	—	—	—	—	—

4. Solution method

The ground-breaking Hybrid NSGAII-MOPSO-TOPSIS algorithm, as elucidated in our seminal contributions [59,60,61,25,22,23,21,29], serves as a cornerstone in addressing the complexities of collaborative planning models. This algorithm operates with precision, concurrently minimizing three pivotal objective functions: total annual expected cost, power loss, and voltage deviation. It plays a crucial role in determining the optimal number, placement, and sizing of wind DGs, PV DGs, PEV-PLs, and SBESSs, as well as in configuring the sizes and monthly transportation schedules of MBESSs. Additionally, it orchestrates the intricate, hourly scenario-based scheduling of the charging and discharging profiles for PEVs, PEV-PLs, SBESSs, and MBESSs with remarkable finesse.

The hybridization of NSGA-II and MOPSO provides substantial advantages in solving complex multi-objective optimization problems, particularly in energy systems planning and operation. This hybrid approach capitalizes on the complementary strengths of its components: the robust exploration and diversity-preserving capabilities of NSGA-II and the rapid convergence of MOPSO. By combining these methodologies, the algorithm achieves a more effective balance between exploration and exploitation, leading to a comprehensive and well-distributed Pareto front. In a related previous work, referenced as [60], the authors compared the performance of NSGA-II, MOPSO, and a hybrid NSGA-II-

Table 12
Monthly optimal locations and sizes of MBESSs considering case 5.

Optimal	Month	Optimal locations and sizes of MBESSs considering case 5		
		MBESS1	MBESS2	MBESS3
Bus location	Jan	21	50	32
	Feb	64	27	46
	Mar	59	24	15
	Apr	19	49	3
	May	51	12	16
	Jun	43	49	34
	Jul	22	46	27
	Aug	35	6	25
	Sep	58	43	56
	Oct	41	47	23
	Nov	29	13	69
	Dec	64	42	2
Size	Number of units	1	5	5
	MWh	0.09	0.45	0.45
	MVA	0.018	0.09	0.09

MOPSO approach. The results highlighted the superiority of the hybrid method in achieving the lowest objective values compared to the other methods.

Importantly, the hybridization process does not rely on weighted objective functions, which can introduce subjective bias and demand

Table 13

The obtained optimal capacities for various allocated sources across different cases.

Total capacities	Case 2	Case 3	Case 4	Case 5
Wind	2 MVA	2.2MVA	2.15	1.55
PV	1.1MVA	1.35MVA	1.25	1.75
RESs	3.1MVA	3.55MVA	3.40	3.30
PEVs	146	206.00	212.00	159.00
PEV-PL	6.132MWh, 1.9272MVA	8.652MWh, 2.7192MVA	8.904MWh, 2.7984MVA	6.678MWh, 2.0988MVA
SBESS	0.00	1.17MWh, 0.234MVA	0.00	0.63MWh,0.126MVA
MBESS	0.00	0.00	0.81MWh, 0.162MVA	0.99MWh, 0.198 MVA
BESS	0.00	1.17MWh, 0.234MVA	0.81MWh, 0.162MVA	1.62MWh, 0.324MVA

prior knowledge of the importance of objectives. Instead, it operates purely on Pareto dominance principles, ensuring that solutions are evaluated based on their trade-offs without prioritizing any single objective upfront. This Pareto-based approach provides decision-makers with a diverse set of optimal solutions, offering greater flexibility and transparency when selecting trade-offs that align with specific priorities. Such a methodology is particularly beneficial in energy optimization problems, where conflicting objectives—such as those proposed in this model—must be addressed simultaneously. By hybridizing NSGA-II and MOPSO, the algorithm enhances solution quality, convergence speed, and decision-making robustness in multi-objective optimization contexts. For a detailed step-by-step procedural guide to the successful implementation of the proposed NSGAII-MOPSO-TOPSIS method, readers are encouraged to refer to our recent published works in [59,60,61,25,22,23,21,29].

5. Results and discussions

To rigorously evaluate the efficacy of our proposed planning model, we employ the 69-bus DS, depicted in Fig. 2, as our testing ground.

Table 14

The obtained optimal proposed objective functions across various cases.

Cases	Optimal values for the proposed short-term objective functions (Minimize)					Green energy penetration (Maximize)	
	Total Investment and Maintenance cost (\$/year)	Total Operation cost (\$/year)	Total expected cost (\$/year)	Total expected power loss (MVA)	Total expected voltage deviation (pu)	Number of PEVs	RESs (MVA)
Case 1	0.00	3506689.23	3506689.23	1681.85	12821.77	0.00	0.00
Case 2	506383.45	2177301.12	2683684.57	1094.14	7993.39	146.00	3.10
Case 3	675366.64	2040242.22	2715608.86	998.02	7953.56	206.00	3.55
Case 4	659975.52	2086292.92	2746268.44	1138.87	8092.40	212.00	3.40
Case 5	581861.27	2373002.61	2954863.88	1240.90	8717.45	159.00	3.30
% R w.r.t. Case 1	Case 2	NA	37.91	23.47	34.94	37.66	NA
	Case 3	NA	41.82	22.56	40.66	37.97	NA
	Case 4	NA	40.51	21.68	32.28	36.89	NA
	Case 5	NA	32.33	15.74	26.22	32.01	NA
% R w.r.t. Case 2	Case 3	-33.37	6.29	-1.19	8.78	0.50	29.13
	Case 4	-30.33	4.18	-2.33	-4.09	-1.24	31.13
	Case 5	-14.91	-8.99	-10.10	-13.41	-9.06	8.18
% R w.r.t. Case 3	Case 4	2.28	-2.26	-1.13	-14.11	-1.75	2.83
	Case 5	13.85	-16.31	-8.81	-24.34	-9.60	-29.56
% R w.r.t. Case 4	Case 5	11.84	-13.74	-7.60	-8.96	-7.72	-33.33
							-3.03

R: Reduction; w.r.t: with respect to.

Delving into the minutiae of technical and economic parameters, Tables 1, 2, 3, and 4 meticulously outline the input economic and technical details regarding the system under study, wind DGs and PV DGs, PEV-PLs and PEVs, and SBESSs and MBESSs. Moreover, Table 5 presents PEV's probability distribution parameters.

5.1. Input profiles

Figs. 3, 4, 5, and 6 provide an exhaustive portrayal of monthly profiles for stochastic variables, including load power, wind speed, solar irradiation, and energy prices across scenarios A and B. These profiles are intricately constructed using MCS and BRM, harnessing a seven-year dataset meticulously curated from sources [63,64], and [65]. As detailed in Section 2, Table 6 displays the months during which uncertain variables, such as load, wind, PV, and energy prices, reached their maximum and minimum levels.

5.2. Case of studies

In this study, we undertake a rigorous examination and analysis of five distinct configurations via simulation, classified as follows:

Case 1: Base case.

Case 2: Hybrid solver + model (Wind DGs, PV DGs, and PEV-PL).

Case 3: Hybrid solver + model (Wind DGs, PV DGs, PEV-PL, and SBESS).

Case 4: Hybrid solver + model (Wind DGs, PV DGs, PEV-PL, and MBESS).

Case 5: Hybrid solver + model (Wind DGs, PV DGs, PEV-PL, SBESS, and MBESS).

5.3. Optimal results

Tables 5 through 12 provide a comprehensive numerical depiction of the optimal locations and capacities for DERs, encompassing Wind DGs, PV DGs, PEV-PL infrastructures, SBESSs, and MBESSs, along with the monthly relocation schedule. They outline these specifics across five distinct configurations. Furthermore, Table 13 meticulously details each

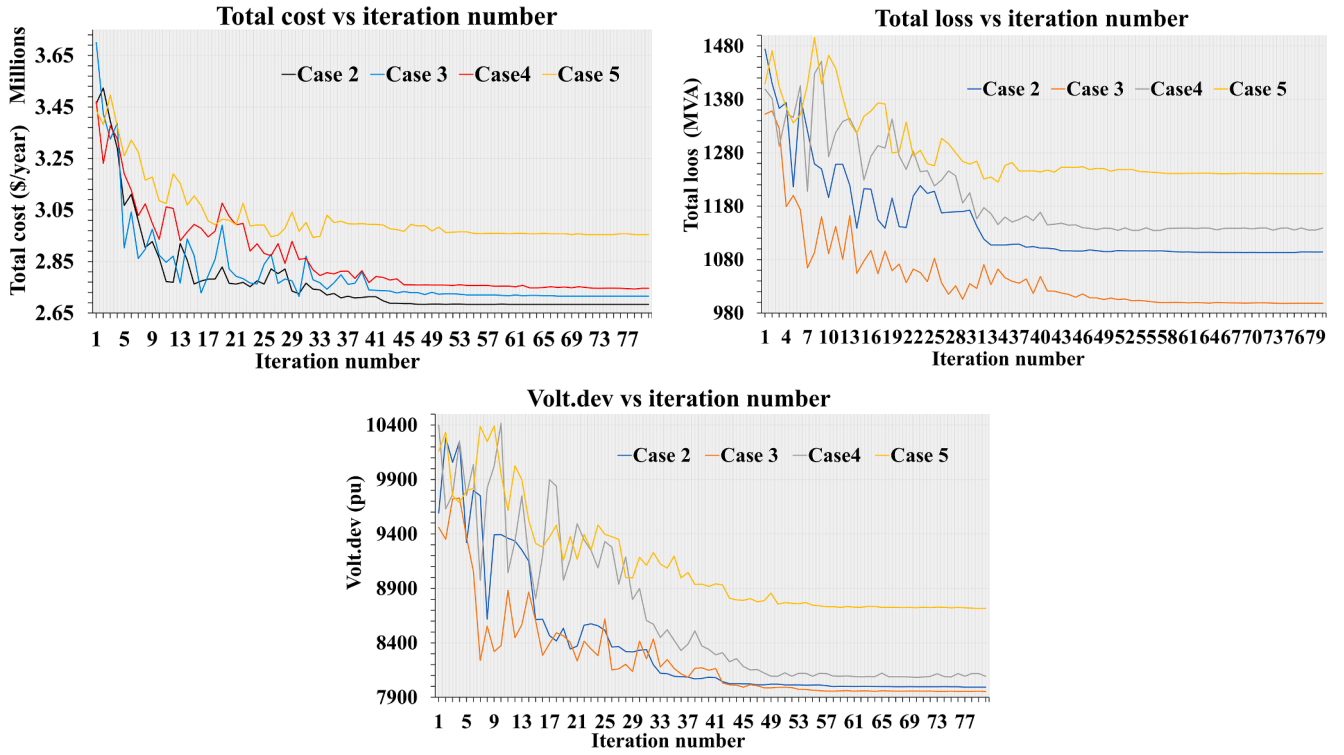


Fig. 7. The convergence characteristic of the proposed objective functions using the hybrid NSGAI-MOPSO-TOPSIS.

case's total optimal capacity allocation for Wind DGs, PV DGs, RESs, PEVs, PEV-PLs, SBESSs, and MBESSs.

In Case 2, the solver of the proposed model optimally allocated 10 Wind DGs, 8 PV DGs, and 4 PEV-PLs across various sites and sizes, as delineated in Table 7. The optimal capacities for Wind DGs and PV DGs ranged from 0.1MVA to 0.25MVA and 0.05MVA to 0.25MVA, respectively. The PEV-PL infrastructure, comprising four allocated sites, can accommodate a fleet of PEVs ranging from 25 PEVs in PEV-PL2 at bus 2 to 53 PEVs in PEV-PL4 at bus 59. In this case and based on the presented optimal results in Table 13, the total deployed capacity for RESs amounted to 3.1MVA. At the same time, the PEV-PLs collectively achieved a total capacity of 6.132MWh and 1.9272MVA.

In Case 3, the optimization solver allocated 10 Wind DGs, 7 PV DGs, 5 PEV-PLs, and 5 SBESSs across diverse sites and sizes, as detailed in Table 8. Wind and PV DGs' optimal capacities ranged from 0.1MVA to 0.25MVA. Additionally, the deployment of 5 SBESSs, with capacities ranging from 0.09MWh and 0.018MVA at bus 24 to 0.36MWh and 0.072MVA at bus 17, was recorded. The fleet size of PEVs varied, with a minimum of 1 PEV in PEV-PL1 located at bus 33 and a maximum of 138 PEVs in PEV-PL5 located at bus 46.

When comparing Case 3 to Case 2, as evidenced by the total capacities outlined in Table 13, the incorporation of 5 SBESSs with a combined installed capacity of 1.17MWh and 0.234MVA led to an increase in the integration of RESs from 3.1MVA to 3.55MVA. Additionally, it raised the penetration level of PEVs from 146 PEVs in Case 2 to 206 PEVs in Case 3. This comparison forms a foundational point, elaborated upon and stated conclusively in the conclusion section.

Similarly, in Case 4, the optimization solver of the proposed model allocated 10 Wind DGs, 9 PV DGs, and 5 PEV-PLs across various sites and sizes, as outlined in Table 9. Wind and PV DGs' optimal capacities ranged from 0.1MVA to 0.25MVA. The fleet size of PEVs varied, with a minimum of 3 PEVs in PEV-PL1 located at bus 44 and a maximum of 145 PEVs in PEV-PL5 located at bus 47.

Furthermore, Table 10 details the monthly bus relocation schedule for the four deployed MBESSs. This schedule highlights the adaptable nature of MBESS deployment, with monthly adjustments in bus

locations to accommodate weather conditions and load level fluctuations. Additionally, the deployment of 4 MBESSs, with capacities ranging from 0.09MWh and 0.018MVA for MBESS1 at various monthly bus locations to 0.36MWh and 0.072MVA for MBESS4 at various monthly bus locations, was recorded.

When comparing Case 4 to Case 3, as indicated by the total capacities outlined in Table 13, the inclusion of 4 MBESSs resulted in a combined installed capacity of 0.81MWh and 0.162MVA, significantly lower than the deployment of 5 SBESSs (1.17MWh and 0.234MVA) in Case 3. This translates to a reduced investment cost on storage devices. Moreover, the lower installed capacities of MBESSs in Case 4 led to a decrease in the integration of RESs from 3.55MVA to 3.4MVA compared to the deployment of 5 SBESSs in Case 3. However, deploying MBESSs instead of SBESSs increased the penetration level of PEVs from 206 PEVs to 212 PEVs. This comparison establishes a second conclusion point, further expounded upon and conclusively stated in the conclusion section.

Case 5 introduces a configuration that combines Wind DGs, PV DGs, and PEV-PLs with a hybrid deployment of SBESSs and MBESSs. The optimization solver of the proposed model allocated 10 Wind DGs, 10 PV DGs, and 5 PEV-PLs across various sites and sizes, as delineated in Table 11. Wind DGs were optimized within a range of 0.1MVA to 0.25MVA, while PV DGs varied from 0.05MVA to 0.25MVA. The fleet size of PEVs exhibited variability, with a minimum of 2 PEVs in PEV-PL1 located at bus 52 and a maximum of 82 PEVs in PEV-PL5 located at bus 39.

Furthermore, the deployment of 3 SBESSs was recorded, with capacities ranging from 0.09MWh and 0.018MVA at bus 46 to 0.36MWh and 0.072MVA at bus 2. Additionally, the monthly bus relocation schedule for the three deployed MBESSs is detailed in Table 12. Moreover, the deployment of 3 MBESSs was observed, with capacities ranging from 0.09MWh and 0.018MVA for MBESS1 at various monthly bus locations to 0.45MWh and 0.09MVA for MBESS2 and MBESS3 at various monthly bus locations.

Comparing Case 5 to Case 2, as evidenced in Table 13, the hybrid deployment of 3 SBESSs and 3 MBESSs increased the total deployment of RESs from 3.1MVA to 3.3MVA and raised the penetration level of PEVs

Table 15

The obtained optimal monthly costs for different sources across various cases 1, 2, and 3.

Case	Sources	Cost (\$/month)	Month												Total
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
1	DSS	Oper	3071 61.2	2691 81.1	2083 08.6	1952 24.9	1690 92.8	2306 64.3	3307 11.3	3158 71.0	1979 09.9	1845 67.0	2331 66.7	2495 16.4	2891 375.1
		Em	4956 2.5	4798 0.3	4771 4.7	4477 4.7	4621 3.3	5632 3.1	6306 9.2	6355 3.1	4998 1.6	4756 0.4	4935 2.7	4922 8.5	6153 14.1
	Total		3567 23.7	3171 61.4	2560 23.3	2399 99.6	2153 06.1	2869 87.4	3937 80.5	3794 24.1	2478 91.5	2321 27.4	2825 19.4	2987 44.9	3506 689.2
2	DSS	Oper	1867 23.8	1909 20.4	1622 51.7	7201 0.2	9948 4.0	1275 81.5	2540 16.1	2320 84.6	1144 34.0	1337 75.3	1187 06.4	1239 74.6	1815 962.7
		Em	3022 8.9	3241 7.0	3569 9.5	1562 6.2	2738 3.4	2923 6.8	4710 3.6	4767 0.8	2915 2.8	3290 1.0	2398 7.6	2418 8.0	3755 95.6
	Total		2169 52.7	2233 37.3	1979 51.2	8763 6.4	1268 67.5	1568 18.3	3011 19.7	2797 55.3	1435 86.9	1666 76.3	1426 94.0	1481 62.6	2191 558.3
3	W	Oper	1934 .1	1656 .0	1011 .8	2990 .4	1657 .8	2444 .7	1283 .0	1169 .4	1810 .4	1571 .8	2608 .3	2573 .8	2271 1.4
		PV	84.5	33.0	235. 6	71.7	276. 4	288. 3	291. 2	360. 9	294. 8	31.4	52.8	37.9	2058. 6
	PEV -PL	Oper	110. 9	106. 8	114. 9	101. 2	111. 6	104. 2	105. 0	111. 2	110. 0	112. 0	108. 9	109. 7	1306. 5
		Energy buying	2550 .3	2114 .4	1979 .9	1868 .1	1635 .0	1567 .7	2121 .7	2225 .3	1805 .7	1662 .6	1897 .6	2330 .9	2375 9.3
		Rev	7508 .8	6878 .4	5474 .2	4552 .0	4250 .0	4564 .6	5296 .1	5261 .7	4125 .9	4544 .8	5816 .0	5820 .4	6409 2.9
		Total benefits	4847 .6	4657 .2	3379 .4	2582 .7	2503 .3	2892 .7	3069 .4	2925 .2	2210 .1	2770 .2	3809 .5	3379 .8	3902 7.1
	Total		2141 23.8	2203 69.2	1958 19.3	8811 5.8	1262 98.3	1566 58.6	2996 24.5	2783 60.4	1434 82.0	1655 09.2	1415 45.7	1473 94.5	2177 301.1
	DSS	Oper	1763 56.7	1856 34.5	1579 80.8	6078 9.0	9246 9.7	1168 10.3	2447 30.4	2215 44.2	1050 26.0	1299 93.8	1093 20.6	1133 88.0	1714 044.1
		Em	2870 7.5	3139 2.2	3460 0.6	1301 6.9	2556 6.5	2655 3.2	4542 9.5	4589 1.5	2705 8.2	3182 3.6	2178 6.4	2216 5.5	3539 91.7
		Total	2050 64.2	2170 26.7	1925 81.4	7380 5.9	1180 36.1	1433 63.6	2901 59.9	2674 35.8	1320 84.2	1618 17.4	1311 07.0	1355 53.5	2068 035.7
	W	Oper	2127 .5	1821 .6	1113 .0	3289 .4	1823 .6	2689 .1	1411 .3	1286 .4	1991 .4	1728 .9	2869 .1	2831 .1	2498 2.5
	PV	Oper	103. 8	40.5	289. 2	88.0	339. 2	353. 9	357. 4	442. 9	361. 8	38.5	64.8	46.5	2526. 5
3	PEV -PL	Oper	156. 7	158. 2	159. 8	144. 1	153. 7	149. 1	145. 8	157. 3	150. 1	148. 2	154. 5	154. 4	1831. 9
		Energy buying	3528 .4	3207 .4	2912 .5	2611 .9	2148 .0	2334 .7	2972 .1	3181 .7	2377 .4	2109 .9	2679 .8	3120 .3	3318 4.0
		Rev	1036 9.9	1032 6.9	7544 .9	6411 .8	5990 .3	6556 .7	7517 .9	7515 .7	5730 .9	6191 .8	8392 .4	8515 .0	9106 4.0
		Total benefits	6684 .8	6961 .3	4472 .6	3655 .8	3688 .5	4073 .0	4400 .0	4176 .7	3203 .5	3933 .7	5558 .1	5240 .3	5604 8.2
	SB	Opera	64.5	64.3	63.8	63.8	63.6	61.3	61.0	62.7	58.7	63.5	54.3	64.0	745.7
	Total		2006 75.1	2119 91.8	1895 74.9	7359 1.4	1165 74.1	1423 94.9	2875 89.6	2650 51.1	1312 92.6	1597 14.7	1285 37.1	1332 54.9	2040 242.2

Oper: Operation; Em: Emission; W: Wind; SB: SBESS; Red color: Maximum values; Green color: Minimum values; Yellow color: Total values.

from 146 PEVs to 159 PEVs. Across all cases, a consistent enhancement in the penetration level of green energy was achieved by increasing the capacity for RESs and PEVs penetration. This enhancement was observed, whether considering the deployment of SBESSs alone, MBESSs alone, or a hybrid SBESS-MBESS system. This comparison constitutes a third key finding, which will be further developed and conclusively articulated in the conclusion section of the study.

Comparing Case 5 to Case 2 and Case 3, as shown in Table 13, the deployed capacity of PEV-PLs in Case 5 was the lowest compared to Case 3 and Case 4, primarily due to the reduced number of penetrated PEVs, totalling 159 PEVs compared to Case 3 (206 PEVs) and Case 4 (212 PEVs). Consequently, the investment required for PEV-PL infrastructure is considerably lower in Case 5. Additionally, the penetration level of RESs decreased from 3.55MVA in Case 3 and 3.4MVA in Case 4 to

3.3MVA in Case 5. This indicates a lower investment required for RESs than Case 4 and 5. However, the deployment of a hybrid system comprising 3 SBESSs and 3 MBESSs increased the total capacity of storage devices from (1.17MWh, 0.234MVA) in Case 3 and (0.81MWh, 0.162MVA) in Case 4 to 0.99MWh and 0.198MVA in Case 5, necessitating a more significant investment in storage devices.

Consequently, despite the hybrid mode not demonstrating superior capability in enhancing the penetration level of RESs and PEVs compared to SBESSs alone or MBESSs alone, considering the short-term planning of one year, the reduction in the annual investment cost could be a significant economic advantage for long-term planning. This is evidenced by the total yearly investment and maintenance cost of the system in Case 5, which recorded its lowest value at 581,861.27\$/year compared to Case 3 (675,366.64\$/year) and Case 4 (659,975.52

Table 16

The obtained optimal monthly costs for different sources across various cases 4 and 5.

Case	Sources	Cost (\$/month)	Month												Total
			Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
4	DSS	Oper	1798 72.3	1873 59.1	1597 52.0	6408 1.7	9506 7.7	1202 12.8	2481 30.8	2255 50.0	1082 34.5	1312 58.4	1121 65.9	1167 83.4	1748 468.4
		Em	2950 1.2	3181 1.6	3507 6.4	1407 5.7	2635 8.9	2779 9.4	4622 1.2	4678 8.2	2803 4.8	3222 8.9	2281 0.9	2315 1.7	3638 59.0
		Total	2093 73.5	2191 70.7	1948 28.4	7815 7.4	1214 26.6	1480 12.2	2943 52.0	2723 38.1	1362 69.3	1634 87.3	1349 76.8	1399 35.1	2112 327.3
	W	Oper	2079 .1	1780 .2	1087 .7	3214 .7	1782 .1	2628 .0	1379 .2	1257 .1	1946 .1	1689 .6	2803 .9	2766 .8	2441 4.7
	PV	Oper	96.1	37.5	267. 8	81.5	314. 1	327. 6	330. 9	410. 1	335. 0	35.6	60.0	43.1	2339. 3
	PEV-PL	Oper	163. 1	161. 2	164. 9	148. 8	160. 6	152. 7	151. 4	161. 3	155. 0	151. 8	159. 4	159. 2	1889. 2
		Energy buying	3598 .0	3350 .3	2993 .6	2703 .3	2248 .3	2446 .0	3122 .4	3092 .6	2414 .0	2195 .9	2743 .7	3136 .8	3404 5.0
		Rev	1086 0.1	1043 3.4	7764 .3	6666 .9	6286 .5	6658 .2	7648 .4	7800 .9	5954 .6	6327 .2	8622 .1	8754 .3	9377 6.9
		Total benefits	7099 .0	6921 .9	4605 .9	3814 .8	3877 .7	4059 .5	4374 .6	4547 .0	3385 .6	3979 .5	5719 .0	5458 .4	5784 2.7
	MB	Opera	44.8	44.5	44.2	44.2	44.1	42.5	42.2	43.4	40.6	43.9	37.6	44.5	516.6
		Transp	0.0	408. 8	413. 3	410. 0	413. 3	414. 3	415. 0	406. 0	409. 3	414. 0	413. 8	420. 3	4537. 8
		Total	44.8	453. 2	457. 5	454. 2	457. 3	456. 7	457. 2	449. 4	449. 9	457. 9	451. 3	464. 7	5054. 3
	Total		2044 94.5	2145 19.7	1920 35.5	7809 3.0	1201 02.4	1473 65.1	2921 44.8	2699 07.8	1356 14.7	1616 91.0	1325 73.1	1377 51.4	2086 292.9
	DSS	Opera	2114 18.9	2083 40.9	1646 89.7	9755 1.7	1056 51.4	1390 13.9	2544 91.2	2312 47.6	1208 33.7	1449 16.8	1437 20.3	1518 06.9	1973 682.9
		Em	3450 3.2	3602 1.0	3657 9.2	2206 0.0	2946 9.7	3324 3.6	4839 2.8	4827 0.4	3157 8.5	3627 4.4	2980 0.1	3011 8.5	4163 11.2
		Total	2459 22.1	2443 61.8	2012 68.9	1196 11.7	1351 21.1	1722 57.5	3028 83.9	2795 17.9	1524 12.2	1811 91.2	1735 20.4	1819 25.4	2389 994.1
5	W	Oper	1498 .9	1283 .4	784. 2	2317 .5	1284 .8	1894 .6	994. 3	906. 3	1403 .0	1218 .1	2021 .4	1994 .7	1760 1.3
	PV	Oper	134. 5	52.6	374. 9	114. 1	439. 7	458. 7	463. 3	574. 1	469. 0	49.9	84.0	60.3	3275. 1
	PEV-PL	Oper	125. 8	121. 2	125. 3	113. 5	119. 1	120. 6	118. 1	119. 9	119. 8	119. 5	125. 0	118. 5	1446. 4
		Energy buying	2698 .6	2586 .4	2294 .7	2039 .2	1754 .4	1965 .8	2355 .2	2320 .8	1894 .3	1789 .7	2330 .8	2439 .3	2646 9.1
		Rev	8471 .2	7530 .6	5943 .9	4952 .6	4574 .2	4944 .7	6004 .4	5750 .0	4572 .9	4822 .1	6375 .4	6275 .6	7021 7.6
		Total benefits	5646 .8	4822 .9	3523 .9	2799 .9	2700 .7	2858 .3	3531 .1	3309 .3	2558 .8	2912 .9	3919 .5	3717 .8	4230 2.1
	SB	Oper	34.9	34.9	34.4	34.5	34.0	33.0	32.9	33.8	31.6	34.4	29.2	34.6	402.0
	MB	Oper	54.8	54.8	53.9	54.0	53.4	51.8	51.6	53.1	49.5	54.0	46.0	54.4	631.2
		Transp	0.0	311. 3	303. 3	310. 3	309. 0	307. 0	313. 8	308. 0	308. 0	310. 8	310. 0	309. 8	3401. 0
		Total	54.8	366. 0	357. 1	364. 3	362. 4	358. 8	365. 4	361. 1	357. 5	364. 8	356. 0	364. 1	4032. 2
	Total		2419 98.3	2412 75.7	1992 95.6	1196 42.2	1345 41.2	1721 44.3	3012 08.7	2780 83.9	1521 14.5	1799 45.4	1720 91.5	1806 61.3	2373 002.6

Oper: Operation; Em: Emission; W: Wind; SB: SBESS; MB: MBESS; Transp: Transportation; Red color: Maximum values; Green color: Minimum values; Yellow color: Total values.

\$/year), as depicted in Table 14. This comparison serves as an additional conclusion point, offering further insight. It will be thoroughly discussed and conclusively presented in the concluding remarks section.

Table 14 presents a comprehensive summary of the optimal objective function values achieved, along with the penetration levels of green energy, encompassing RESs, and the fleet size of PEVs considered in the short-term planning spanning one year. The yellow-shaded cells indicate the optimal case based on the values of the various obtained objective

functions. In comparing the five proposed configurations, Case 3 (wind/PV/PEV-PL/SBESS) emerges as the optimal choice, boasting the lowest total expected operation cost, power loss, and voltage deviation, indicative of its efficacy in short-term planning and operation over one year. Case 3 achieved a notable reduction in the total expected operation cost by 41.82 %, 6.29 %, 2.26 %, and 16.31 % when juxtaposed with Cases 1, 2, 4, and 5, resulting in a value of 2,040,242.22\$/year. Additionally, it demonstrated reductions in the total expected power loss by 40.66 %,

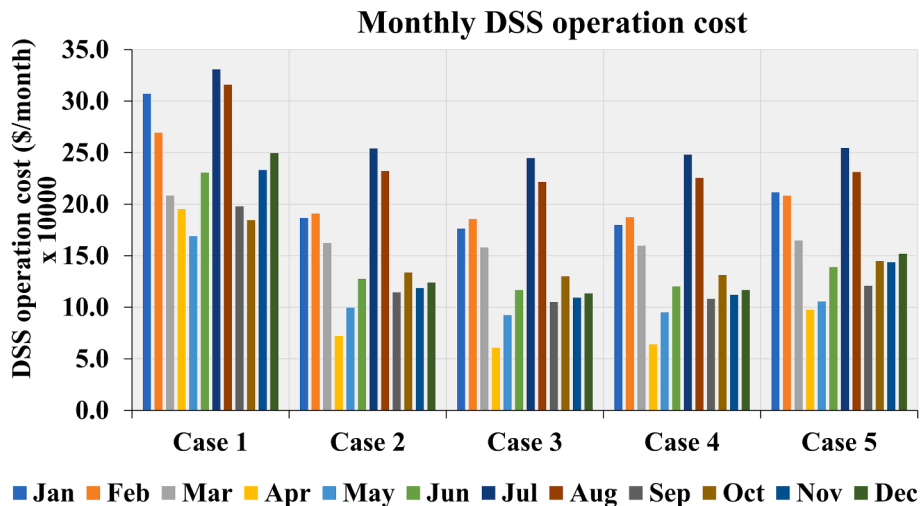


Fig. 8. The obtained optimal monthly DSS operation cost across various cases.

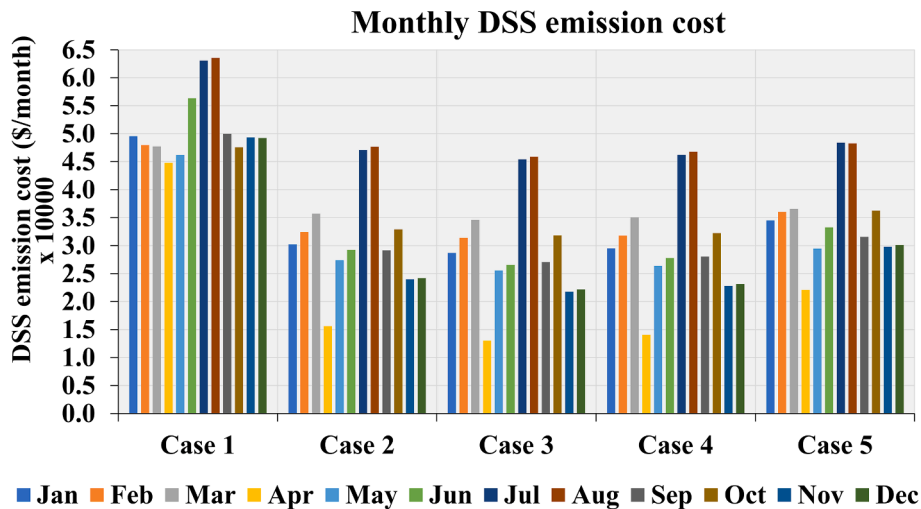


Fig. 9. The obtained optimal monthly DSS emission cost across various cases.

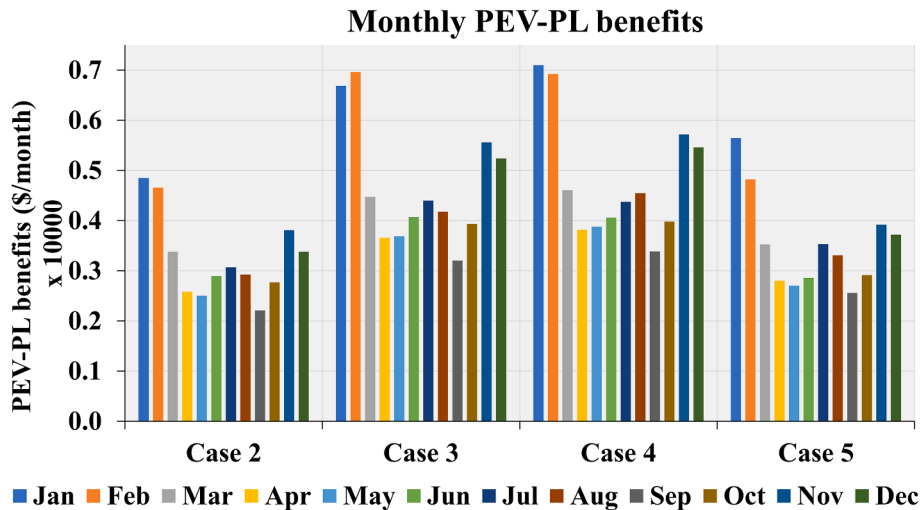


Fig. 10. The obtained optimal monthly PEV-PL's total benefits across various cases.

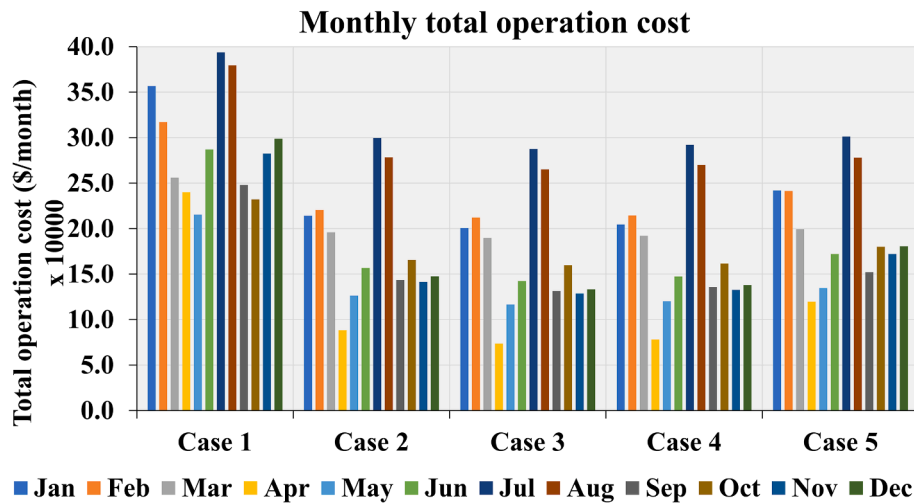


Fig. 11. The obtained optimal monthly total operation cost across various cases.

Table 17

The obtained optimal monthly technical objectives across various cases.

Month	Expected power loss (MVA)					Expected voltage deviation (pu)				
	Case 1	Case 2	Case 3	Case 4	Case 5	Case 1	Case 2	Case 3	Case 4	Case 5
Jan	113.06	66.52	59.67	71.35	82.11	1032.33	602.51	623.32	614.87	714.04
Feb	105.79	83.68	81.11	84.96	88.29	999.00	706.61	703.40	718.73	759.70
Mar	106.38	80.84	77.59	80.01	81.77	992.99	752.63	746.67	746.12	747.45
Apr	99.61	48.99	40.77	56.51	66.53	930.87	363.93	355.77	404.51	492.80
May	112.34	70.58	64.46	71.77	78.29	961.15	592.71	577.72	584.41	612.00
Jun	181.45	97.85	86.29	108.07	124.37	1175.21	648.91	641.91	668.42	715.39
Jul	245.40	183.78	171.40	182.51	192.50	1318.86	996.27	984.50	985.44	1003.34
Aug	243.03	169.74	157.31	170.08	181.56	1328.69	1010.22	997.90	986.09	993.79
Sep	137.58	80.94	72.44	86.33	92.91	1040.81	624.92	618.79	625.22	656.21
Oct	108.41	94.45	90.72	94.69	95.43	989.47	726.96	711.37	750.64	766.47
Nov	118.61	64.03	51.00	70.53	82.89	1027.33	502.66	500.80	519.84	627.59
Dec	110.19	52.75	45.25	62.05	74.25	1025.07	465.05	491.41	488.12	628.68
Total	1681.85	1094.14	998.02	1138.87	1240.90	12821.77	7993.39	7953.56	8092.40	8717.45

Red color: Maximum values; Green color: Minimum values.

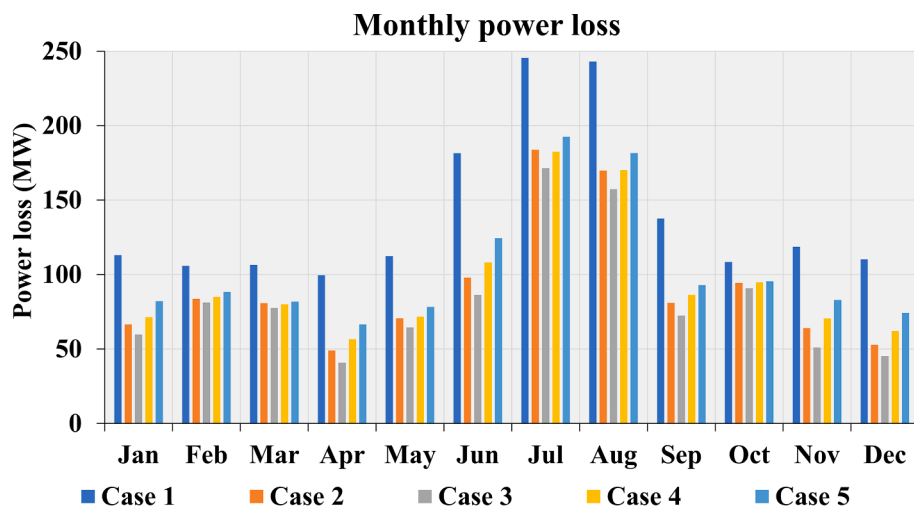


Fig. 12. The obtained optimal monthly total power loss across various cases.

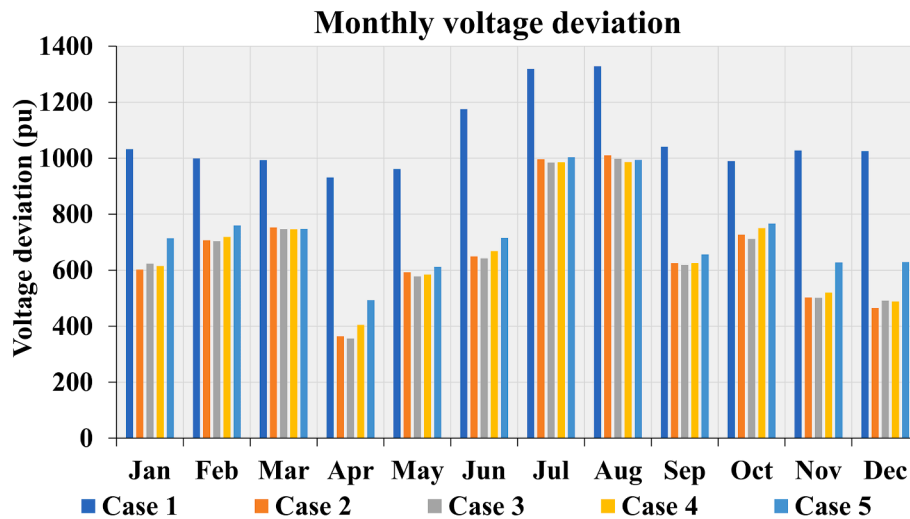


Fig. 13. The obtained optimal monthly voltage deviation across various cases.

8.78 %, 14.11 %, and 24.34 %, and in the total expected voltage deviation by 37.97 %, 0.5 %, 1.75 %, and 9.60 %, respectively, compared to Cases 1, 2, 4, and 5. These improvements culminated in a total expected power loss of 998.02 MVA and a voltage deviation of 7953.56 pu. Additionally, Case 3 attained the most significant penetration of RESs, boasting a total capacity of 3.55 MVA. This signifies a noteworthy increase of 12.68 %, 4.41 %, and 7.58 % compared to Cases 2, 4, and 5, which possessed capacities of 3.1 MVA, 3.4 MVA, and 3.3 MVA, respectively.

However, Case 4 (wind/PV/PEV-PL/MBESS) achieved the highest integration level of PEVs among all cases, with a total capacity of 212 PEVs distributed among 4 PEV-PLs in the DS. This represents an increase of 31.13 %, 2.83 %, and 33.33 % compared to Cases 2, 3, and 5, respectively.

Furthermore, Cases 2, 3, 4, and 5 exhibit superior objective function values compared to Case 1, underscoring the significance of investing in deploying PEV-PLs and storage devices. Despite the deployment of hybrid SBESS and MBESS alongside wind, PV, and PEV-PLs in Case 5, resulting in the poorest performance in terms of total expected operation cost, power loss, voltage deviation, RESs penetration level, and PEVs integration, it achieves the lowest annual investment cost of 581,861.27 \$/year, representing reductions of 14.91 %, 13.85 %, and 11.84 % compared to Cases 2, 3, and 4, respectively. Such a low annual investment cost may position the hybrid case (wind/PV/PEV-PL/SBESS/MBESS) as the optimal choice for long-term projects spanning ten years.

The methodology proposed in this study adopts a hybrid algorithm, merging NSGAI and MOPSO with TOPSIS. This approach aims to optimize the developed planning model over a short-term span of one year. Incorporating the TOPSIS decision-making algorithm allows for deriving an optimal solution from a set of non-dominated solutions linked to three objective functions. Fig. 7 (a), (b), and (c) portray the convergence dynamics of the hybrid algorithm across Cases 2, 3, 4, and 5, illustrating the progression of each objective function throughout iterations. Throughout the convergence process, fluctuations occur due to the simultaneous minimization of three independent objective functions amidst system uncertainty.

Tables 15 and 16 present the optimal monthly and total operation and emission costs for the DSS and operation costs for the wind and PV DGs. They also feature operation costs, expenses from purchasing energy from PEVs, revenues from selling energy to parked EVs, and the total collected profits for the PEV-PLs. The five proposed configurations also outline operation costs for SBESSs and the operation and transportation costs for MBESSs.

Furthermore, the maximum and minimum values for these cost

parameters are visually highlighted in red and green within Tables 15 and 16, aligning with the data presented in Table 6. For example, the peak of DSS operation costs in July across all five cases corresponds to the period of highest power demand and maximum energy prices. Similarly, the peak of DSS emission costs occurs in July and August, coinciding with periods of elevated power load. Additionally, the maximum availability of wind and solar power was observed in April and August, aligning with peak wind speed and solar irradiation, as indicated in Table 6. Furthermore, the highest energy expenses, revenues, and collected benefits for the PEV-PLs were predominantly recorded in January, reflecting the highest energy prices. Moreover, the maximum operation costs for deployed SBESSs and MBESSs were noted in January, driven by the peak operation, energy expenses, and revenues of PEV-PLs during that month. However, the maximum transportation costs of MBESSs were recorded across different months rather than being specific to a single month.

The numerical optimal economic and environmental results obtained in Tables 15 and 16 are visualized in Figs. 8, 9, 10, and 11 for better presentation. So, according to Tables 15 and 16 and Figs. 8 and 9, the DSS total expected operation cost in Case 3 recorded 1714044.1\$/year which is less than those recorded in Cases 1, 2, 4, and 5 by 40.72 %, 5.612 %, 1.96 %, and 13.15 %. Moreover, the DSS total expected emission cost in Case 3 recorded 353991.7\$/year, less than those recorded in Cases 1, 2, 4, and 5 by 42.47 %, 2.71 %, and 14.97 %. This ensures the superiority of using SBESSs alone over other configurations in decreasing the dependency on DSS and making it more environmentally friendly. This comparative analysis provides an additional conclusion point, offering further insights. It will be thoroughly discussed and conclusively presented in the concluding remarks section.

Fig. 10 illustrates the total monthly benefits accrued by the PEV-PLs, highlighting the advantage of using SBESSs or MBESSs alone over hybrid configurations or systems without storage devices in short-term planning. Notably, significant fleet sizes are achieved—206 PEVs with SBESSs alone and 212 PEVs with MBESSs alone—contributing to this superiority. However, Tables 15 and 16 reveal that Case 3 yields an annual benefit of 56,048.2/year, with an additional 3.2019 % recorded for Case 4, increasing benefits to 57,842.7/year. Thus, deploying MBESSs offers operators of PEV-PLs the opportunity to enhance benefits.

Moreover, as indicated in Fig. 11, all cases featuring PEV-PLs, with or without storage devices (Cases 2, 3, 4, and 5), demonstrate significantly lower operation costs than the base case. Furthermore, Case 3 (SBESSs) and Case 4 (MBESSs) notably outperform Case 5 (hybrid SBESSs-MBESSs) in reducing the total expected operation cost for the proposed short-term planning. A numerical comparison between Case 3 and

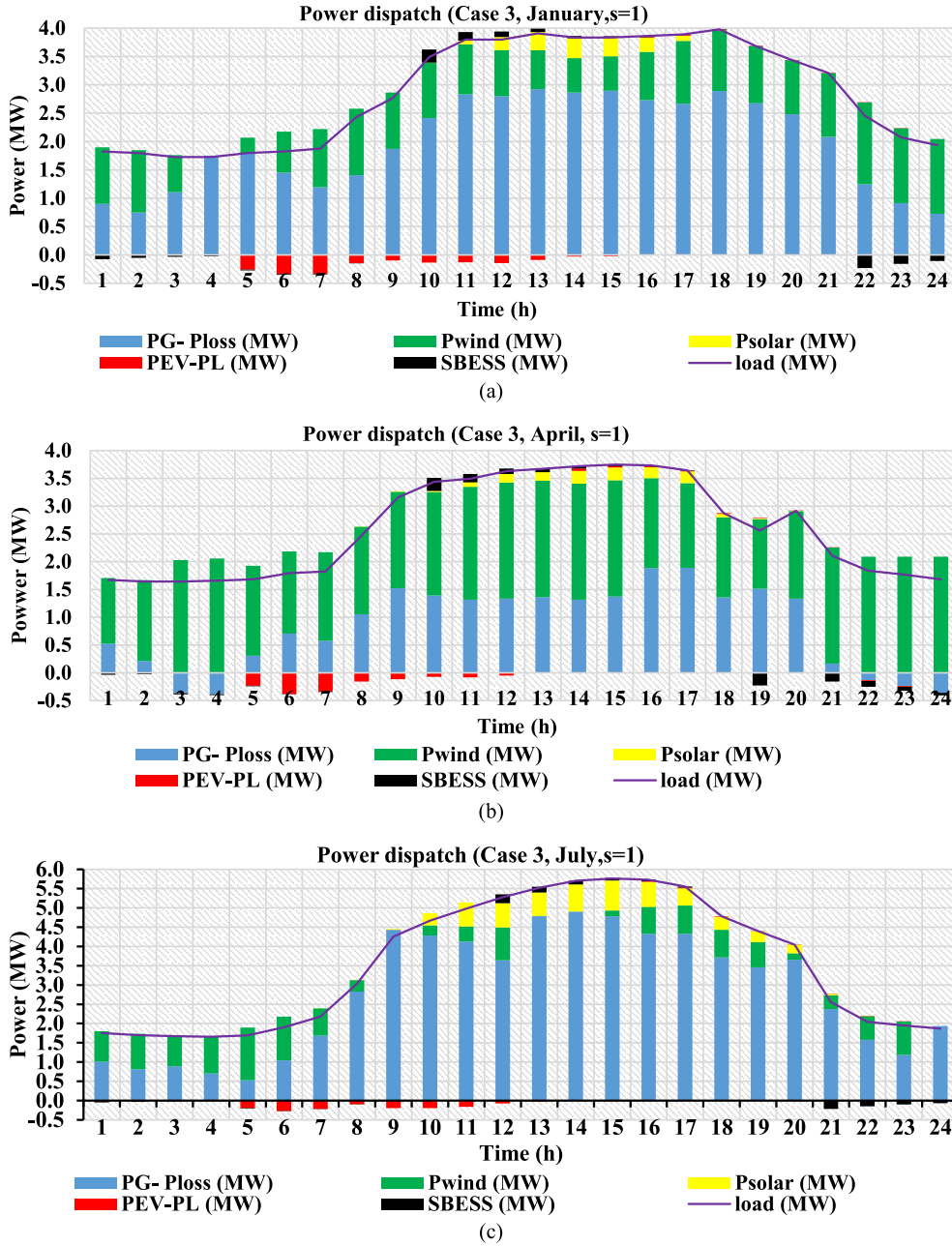


Fig. 14. The obtained optimal hourly power dispatch schedule considering case 3 for various months.

Case 4, drawn from Tables 15 and 16, reveals that employing SBESSs reduce the system's total expected operation cost by 2.257 % compared to using MBESSs. This renders SBESSs superior to MBESSs in reducing total operation costs over a one-year short-term planning horizon.

The proposed model incorporates technical objective functions. Thus, Table 17 presents the obtained optimal monthly and total expected power loss and voltage deviation across various configurations. These results are further visualized in Figs. 12 and 13 for more precise presentation. Accordingly, July and August exhibit the highest power loss and voltage deviation values, attributed to heightened demand during this period. Conversely, April records low power loss and voltage deviation, corresponding to decreased power demand.

Figs. 12 and 13 demonstrate that employing the wind/PV/PEV-PL/SBESS configuration led to the lowest monthly and total expected power loss and voltage deviation. It's important to note that these results are specific to a one-year timeframe and may vary for long-term planning.

The hourly power dispatches for the DSS, wind DGs, PV DGs, PEV-PL systems, and SBESSs are depicted in Fig. 14(a), (b), and (c) for Case 3. This analysis focuses on January, April, and July, strategically selected to capture periods marked by elevated energy prices, maximum wind power availability, and heightened power demand.

In Fig. 14(a), the deployed PEV-PL systems exhibit a high charging mode from 05:00 to 08:00, aligning with a relatively low load power compared to the recorded peak load of 3.97 MW and high wind power availability. Conversely, the SBESSs maintain a much lower charge rate from 01:00 to 08:00, initiated with an initial SOC of 70 %. Notably, the PEV-PLs experience a reduced charge rate from 09:00 to 13:00. Interestingly, the PEV-PLs continue to operate in charging mode even during periods of high and peak load from 10:00 to 13:00, potentially elevating the base peak load. However, optimization strategies direct all SBESSs to discharge their stored energy at a high rate during this timeframe to offset a portion of the developed peak load in conjunction with the available wind and solar energy.

Table 18

The obtained optimal numerical hourly power dispatch schedule in July considering case 5.

Time (h)	load (MW)	(DSS- loss) (MW)	Wind (MW)	PV (MW)	PEV-PL (MW)	SBESS (MW)	MBESS (MW)
1	1.754	1.263	0.557	0.000	0.000	-0.025	-0.040
2	1.699	1.098	0.645	0.000	0.000	-0.017	-0.027
3	1.671	1.136	0.566	0.000	0.000	-0.012	-0.019
4	1.657	0.996	0.681	0.000	0.000	-0.008	-0.013
5	1.692	0.894	0.964	0.000	-0.153	-0.006	-0.009
6	1.902	1.321	0.802	0.000	-0.211	-0.004	-0.006
7	2.177	1.926	0.502	0.001	-0.245	-0.003	-0.004
8	3.032	2.982	0.210	0.007	-0.161	-0.002	-0.003
9	4.259	4.409	0.000	0.030	-0.180	0.000	0.000
10	4.671	4.223	0.183	0.418	-0.152	0.000	0.000
11	4.980	3.999	0.276	0.810	-0.105	0.000	0.000
12	5.275	3.583	0.599	0.815	-0.042	0.125	0.196
13	5.519	4.532	0.005	0.787	-0.013	0.081	0.127
14	5.704	4.636	0.000	0.911	0.022	0.053	0.083
15	5.762	4.562	0.103	1.013	-0.004	0.034	0.054
16	5.732	4.335	0.490	0.843	0.007	0.022	0.035
17	5.554	4.411	0.515	0.584	0.007	0.014	0.023
18	4.783	3.832	0.503	0.444	0.004	0.000	0.000
19	4.399	3.571	0.460	0.365	0.002	0.000	0.000
20	4.045	3.638	0.123	0.283	0.001	0.000	0.000
21	2.555	2.555	0.255	0.045	0.001	-0.117	-0.183
22	2.047	1.819	0.433	0.000	0.000	-0.080	-0.125
23	1.949	1.478	0.611	0.000	0.000	-0.054	-0.086
24	1.866	1.962	0.000	0.000	0.000	-0.037	-0.058

Red: Charging; Green: Discharging; Blue: Floating/not available.

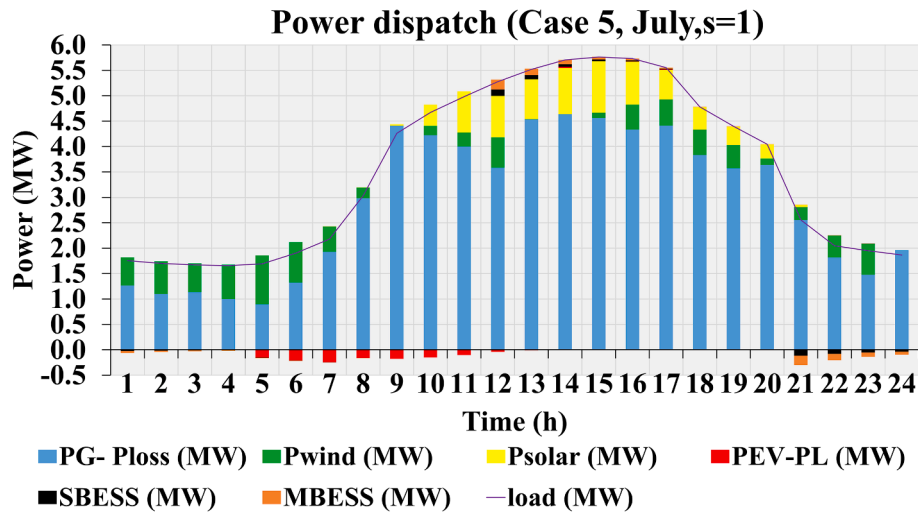


Fig. 15. The optimal hourly power dispatch schedule was obtained in July, considering case 5.

Furthermore, both the SBESSs and PEV-PLs are instructed by the optimization process to discharge their available stored energy at a low rate during high and peak loads from 16:00 to 20:00 by the available RESs, assisting in covering a segment of the basic peak load. Subsequently, due to the high availability of wind energy and low demand for power, the SBESSs are directed to enter a charging mode at a high rate from 22:00 to 24:00 to replenish the loss in stored energy incurred during the previous discharging mode. At the same time, the PEV-PLs utilize their available EV batteries to facilitate the charging of the SBESSs by undergoing a discharging mode until 23:00. Similar observations can be extrapolated from the visualizations provided in Fig. 14 (b) and (c).

Furthermore, the hourly power dispatch for the wind/PV/PEV-PL/

SBESS/MBESS configuration is presented numerically in Table 18 and visually depicted in Fig. 15 for July, considering the first scenario. Notably, the PEV-PLs exhibit zero power exchange during the time intervals of 01:00 to 04:00 and 22:00 to 24:00 due to the unavailability of EVs.

From 01:00 to 8:00, the hybrid SBESS-MBESS systems are engaged in a charging mode, coinciding with low power demand and wind energy availability. Subsequently, these storage systems transition to a floating mode from 09:00 to 11:00, corresponding to moderate demand levels. Interestingly, the PEV-PLs enter a charging mode at 12:00–13:00 and 15:00, aligning with high and peak demand periods. However, guided by the developed optimization algorithm, the hybrid SBESSs-MBESSs switch to a discharging mode from 12:00 to 15:00 to assist in covering

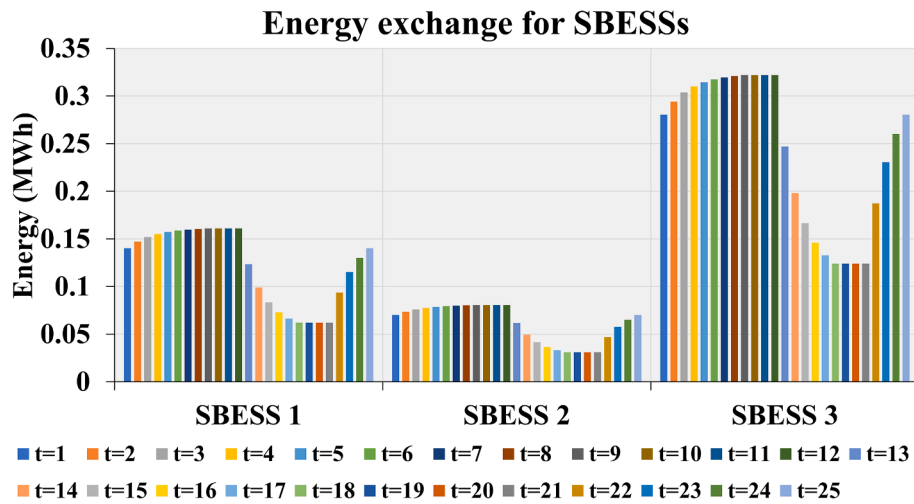


Fig. 16. The obtained optimal energy exchange for each SBESS in July considering case 5.

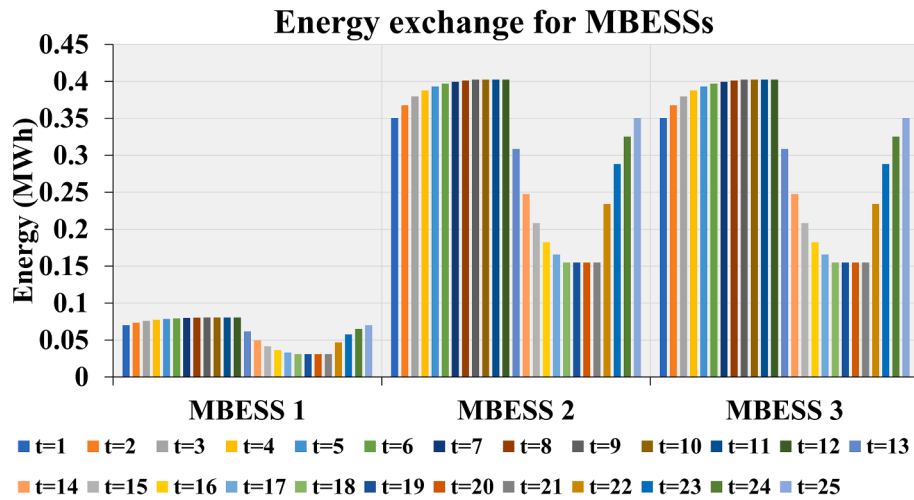


Fig. 17. The obtained optimal energy exchange for each MBESS in July considering case 5.

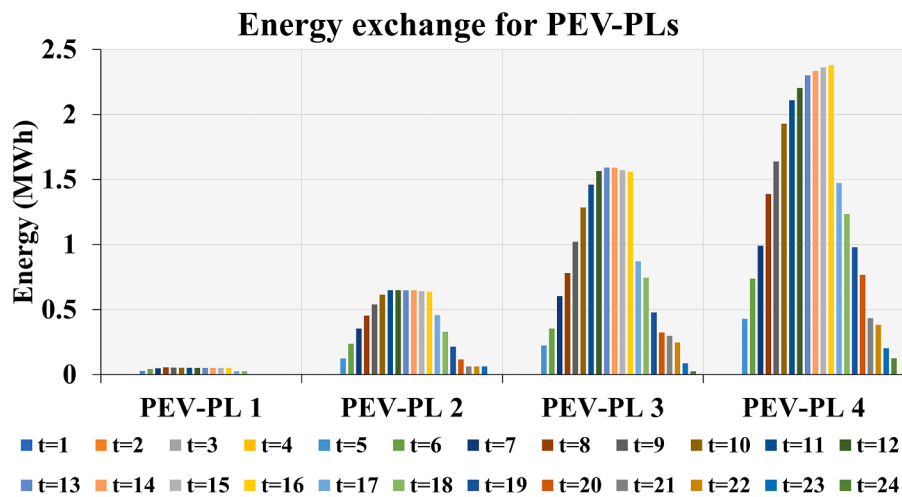


Fig. 18. The obtained optimal energy exchange for each PEV-PL in July considering case 5.

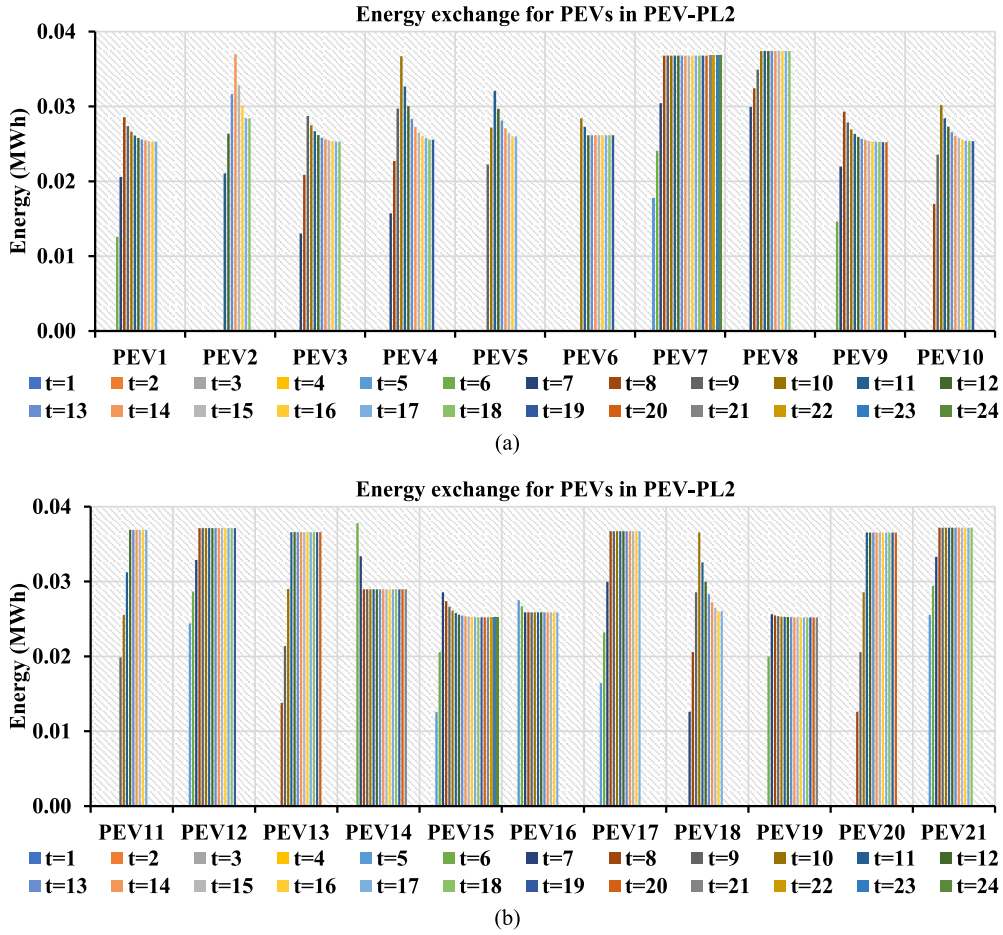


Fig. 19. The obtained optimal energy exchange for each PEV parked in PEV-PL 2 in July, considering case 5.

a portion of the load power, combining the demand from both the PEV-PLs and the primary load demand power by the high availability of solar energy, low wind energy, and the contribution of DSS.

Furthermore, the SBESSs-MBESSs continue in their discharging mode until 17:00 before transitioning to a floating mode from 18:00 to 20:00. Meanwhile, the PEV-PLs persist in contributing to the dispatching of stored energy in the batteries of their parked EVs until 21:00 together with the available wind and solar energy. Between 21:00 and 24:00, the DSS, available wind power, and PEV-PLs collaborate to support the hybrid SBESSs-MBESSs in compensating for their lost energy during the preceding discharging mode.

Furthermore, the energy exchanges for the hybrid SBESSs, MBESSs, and PEV-PLs during charging, discharging, and floating modes are illustrated in Figs. 16, 17, and 18 for Case 5 in July, considering the first scenario. The optimized control strategy developed for the SBESSs and MBESSs in Section 3.7 constrains the number of cycles to 1.5 for each deployed storage device. This strategy is instrumental in extending the lifetime of the storage devices and deferring the need to invest in new installations.

Furthermore, the energy exchange of PEV-PLs depicted in Fig. 18 is contingent upon the designed control strategy outlined in Section 3.4. The devised control strategy for PEV-PLs primarily relies on various parameters of EVs, including arrival time, departure time, arrival energy, and departure energy. Additionally, it considers factors such as current capacity, optimal rates of charge/discharge, availability of RESs, storage devices, load levels, and DSS.

As a result, the optimal control strategy limits the number of charging and discharging cycles of PEV-PLs to only one. For instance, the second PEV-PL deployed at bus 48 has a maximum capacity of 21 PEVs.

The charging and discharging behaviour guided by the designed control strategy for each PEV is illustrated in Fig. 19(a) and (b). This visualization demonstrates that the optimized control strategy restricts each PEV's battery to one cycle, effectively preventing battery degradation, extending its lifespan, and postponing the need for EV owners to invest in purchasing new batteries.

6. Conclusions

This paper presents a short-term optimization model integrating RDGs, such as wind and PV DGs, PEV-PLs, SBESSs, and MBESSs, aiming to enhance green energy and PEV penetration in the DS. The model optimizes decision variables, including RDG location and size, and coordinates optimal charging and discharging strategies for SBESS-MBESS-PEV and monthly relocation schedules for MBESSs. The objective is to minimize annual costs, power loss, and voltage deviation. Validation is performed on a 69-bus DS across five different configurations. The study's findings can be summarized as follows, with the caveat that these results are specific to a one-year timeframe and may vary for long-term planning:

- *Comparing the configuration of (Wind/PV/PEV-PL) without and with SBESSs:* The incorporation of 5 SBESSs with a combined installed capacity of 1.17MWh and 0.234MVA led to an increase in the integration of RESs from 3.1MVA to 3.55MVA and the penetration level of PEVs from 146 PEVs to 206 PEVs.
- *Comparing the configuration of (Wind/PV/PEV-PL) with SBESSs or with MBESSs:* The incorporation of MBESSs alone recorded a combined installed capacity much lower than the deployment of SBESSs alone,

resulting in a much lower investment cost on storage devices. Moreover, such lower installed capacities of MBESSs led to a reduction in the integration of RESs from 3.55MVA in the case of SBESSs alone to 3.4MVA. However, it raised the penetration level of PEVs from 206 PEVs to 212 PEVs.

- *Comparing the configuration of (Wind/PV/PEV-PL) with SBESSs or with MBESSs or with hybrid SBESS-MBESS:* In all cases considering the deployment of SBESSs alone, MBESSs alone, or a hybrid SBESS-MBESS system, there was a consistent enhancement in the penetration level of green energy, achieved by increasing RESs and PEVs penetration capacity, compared to systems without any BESS. Furthermore, Cases 2, 3, 4, and 5 exhibit superior objective function values compared to the base case, underscoring the significance of investing in deploying PEV-PLs and storage devices.
- *Comparing between all configurations:* Wind/PV/PEV-PL/SBESS (Case 3) emerges as the optimal choice, significantly reducing operation costs, power loss, and voltage deviation compared to Cases 1, 2, 4, and 5. Additionally, it achieves the highest RES penetration at 3.55 MVA, surpassing Cases 2, 4, and 5 by 12.68 %, 4.41 %, and 7.58 %, respectively.
- *Comparing the configuration of (Wind/PV/PEV-PL) with SBESSs or with MBESSs or with hybrid SBESS-MBESS:* Despite the hybrid mode (SBESSs-MBESSs) not demonstrating superior capability in minimizing objective functions and enhancing the penetration level of RESs and PEVs compared to SBESSs alone or MBESSs alone considering the short-term planning of one year, the reduction in the annual system's total investment and maintenance cost could be a significant economic advantage for long-term planning.
- *Comparing (wind/PV/PEV-PL/SBESS) with other configurations:* Opting for SBESSs alone, as opposed to either MBESSs alone or a hybrid SBESS-MBESS configuration, ensures superior performance in reducing reliance on the DSS and promoting environmental friendliness considering short term planning.
- *Increasing benefits of PEV-PL's operators:* Utilizing either SBESSs or MBESSs alone yields superior benefits for PEV-PLs compared to hybrid configurations or systems lacking storage devices. Additionally, deploying MBESSs could increase operator benefits by 3.2 % compared to SBESSs.
- *Charging/discharging control strategy of SBESSs and MBESSs:* The optimized control strategy limits each deployed storage device to 1.5 cycles, effectively extending their lifespan and postponing the necessity for new installations.
- *Charging/discharging control strategy of EVs:* The optimized control strategy restricts each PEV's battery to one cycle, thereby preventing battery degradation, prolonging its lifespan, and delaying the need for EV owners to invest in purchasing new batteries.

7. Restrictions and future work

While previous literature has emphasized the advantages of integrating SBESS and MBESS to enhance system resilience during contingency events, this paper's proposed planning model—encompassing wind DG, PV DG, PEV-PL, SBESS, and MBESS—was exclusively evaluated under normal operating conditions and for short-term planning spanning one year. This approach aligns with the core objective of conducting a comprehensive examination of diverse system configurations (Cases 1, 2, 3, 4, and 5) to elucidate the roles of SBESS and MBESS in advancing decarbonization efforts within the power and automotive sectors under standard operational circumstances. Looking ahead, several considerations emerge:

- The proposed model is verified solely for short-term planning, covering one year. However, this timeframe may not provide sufficient insight to determine the optimal configuration concerning investment and operational considerations for long-term planning.

Addressing this limitation represents an area for future research and enhancement.

- The authors plan to augment the existing planning model to include scenarios involving high-impact, low-occurrence natural or artificial disasters. This expansion aims to concurrently enhance both power industry decarbonization and system resilience.
- Furthermore, the authors intend to conduct a deeper investigation into the impacts of high penetration levels of PEVs on system resilience.

Declaration of Generative AI and AI-assisted technologies in the writing process

While preparing this work, the authors used ChatGPT 3.5 to edit the English language professionally. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the publication's content.

CRediT authorship contribution statement

Ahmad K. ALAhmad: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Renuga Verayiah:** Writing – review & editing, Supervision, Resources, Conceptualization. **Hussain Shareef:** Writing – review & editing, Validation, Resources, Funding acquisition. **Agileswari Ramasamy:** Resources, Investigation, Conceptualization. **Saleh Ba-swaimi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by UNITEN through the Ministry of Higher Education Malaysia through a Fundamental Research Grant (FRGS) under project code FRGS/1/2020/TK0/UNITEN/02/2, Postdoctoral Fellowships under project code J510050002, and Emirates Centre for Mobility Research (ECMR)UAEUniversity (project title: D Under Grant 12R202.

Data availability

Data will be made available on request.

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